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## The Rapid Adoption of Generative AI

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# The Rapid Adoption of Generative AI\*

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## Abstract

Generative artificial intelligence (AI) is a potentially important new technology, but its impact on the economy depends on the speed and intensity of adoption. This paper reports results from a series of nationally representative U.S. surveys of generative AI use at work and at home. As of late 2024, 45% of the U.S. population age 18-64 uses generative AI. Among employed respondents, 27% used generative AI for work at least once in the previous week: 10% used it every workday, and 17% on some but not all workdays. Relative to each technology's first mass-market product launch, work adoption of generative AI has been as fast as the personal computer (PC), and overall adoption has been faster than either PCs or the internet. Between 1 and 7% of all work hours are currently assisted by generative AI, and respondents report time savings equivalent to 1.4% of total work hours. Potential productivity gains vary widely by industry, and firm climate and policies play an important role in adoption patterns.

**JEL Codes:** J24, O33

**Keywords:** Generative AI, Technology Adoption, Labor Productivity

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# 1 Introduction

Generative artificial intelligence (genAI) has quickly emerged as a potentially important workplace technology. The large language model ChatGPT debuted in November 2022 and, by December 2024, had over 300 million weekly users. Several studies find that genAI boosts productivity in specific cases (Brynjolfsson et al., 2023; Cui et al., 2024; Dell’Acqua et al., 2023; Demirci et al., 2025; Dillon et al., 2025; Kanazawa et al., 2025; Noy and Zhang, 2023; Otis et al., 2024; Peng et al., 2023), and some project genAI will have large macroeconomic effects (Chui et al., 2023; Filippucci et al., 2024). Yet others predict only modest aggregate impacts (Acemoglu, 2024). These differing views largely reflect uncertainty about the extent of genAI adoption today and in the future.<sup>1</sup>

This paper studies the adoption of genAI using data from the first nationally representative U.S. surveys of genAI usage at work and at home. Our data come from the Real-Time Population Survey (RPS), a national online labor market survey of working age adults aged 18-64 that has run since 2020 (Bick and Blandin, 2023). We report the combined results of two surveys fielded in August and November 2024, which collectively included more than 10,000 respondents. These surveys include questions about genAI adoption that we adapt from the Current Population Survey’s Computer and Internet Supplement (CIU), which was introduced in 1984 to document computer (and later, internet) adoption. We also introduce novel questions to measure how intensively individuals use the technology, which tasks genAI assists with, how much time it saves them, and workplace barriers or support for adoption. The resulting microdata will be made publicly available, addressing calls for more research that develops and shares novel data (Ethiraj et al., 2019).

We use this data to address three main questions. First, how many individuals are currently using genAI? Pooling our August and November 2024 surveys, 32% of workers reported using genAI for work: 10% used it every work day, 17% on some but not all work days, and 5% did not use it in the previous week. One third of respondents report using genAI outside of work. Overall, 45% of respondents report using genAI either for work or outside of work. The most commonly used products are ChatGPT (32% of all respondents), Gemini (19%) and embedded products such as Microsoft Copilot (15%).

To benchmark these results, we show that the rate of workplace genAI adoption in 2024, two years after the release of ChatGPT, is similar to the rate of workplace computer adoption in

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<sup>1</sup>Acemoglu et al. (2022) analyze online job vacancies related to AI, but their data predates ChatGPT. Albanesi et al. (2024), Eisfeldt et al. (2024), and Gathmann et al. (2024) compare occupations according to predictions of AI exposure but do not have data on actual adoption. Humlum and Vestergaard (2025) find no meaningful impacts of genAI adoption on employment and earnings in Denmark among workers in 11 occupations that are particularly exposed to genAI.

1984, three years after the release of the IBM PC (the first mass-market computer). The computer and related information technologies induced far-reaching changes in firm organization and management, which led to substantial productivity gains (Brynjolfsson and Hitt, 1996). The similar pace of genAI adoption thus far suggests that it has the potential to be similarly disruptive. Moreover, we show that genAI adoption by workers is proceeding at a faster pace than adoption by firms, highlighting both challenges and opportunities for managers seeking to capitalize on this new technology (Babina et al., 2024; Bonney et al., 2024; Doshi and Moore, 2025; McElheran et al., 2024).<sup>2</sup>

Second, which workers are currently using genAI, and what factors contribute to an individual’s adoption decision? We find substantial variation in genAI adoption across demographic and labor market characteristics. Adoption has been faster for male, younger, more educated, and higher-wage workers. Adoption also differs greatly by occupation, suggesting that differences in job roles and responsibilities are an important determinant of adoption. Consistent with this, we show that recent estimates of genAI task-based exposure are highly predictive of actual adoption (Eloundou et al., 2024; Felten et al., 2021), and that occupations with lower adoption also have higher shares of workers who report that genAI is not useful for their job. These findings contribute to our understanding of how genAI maps onto specific tasks and roles within the firm, which is relevant both for allocating adoption support and adapting firm organization.

We provide evidence that firm climate and policies also appear to play an important role in workers’ adoption choice. A sizable share of workers who do not use genAI cite workplace barriers as a key reason, and this share is particularly large in a few occupations who may deal with sensitive data. Alternatively, workers whose employers encourage them to use genAI are much more likely to do so, even after controlling for other characteristics. This suggests that when job tasks are amenable to genAI adoption, firms can play an active role by promoting usage, offering training, and removing organizational barriers to adoption.

Third, how large are potential productivity gains from genAI? Based on self-reported time savings due to genAI, the average genAI user would need to work 5.2% more hours in order to complete their work without genAI, implying 1.4% time savings across all workers (including non-users). We use a standard production model to estimate potential productivity gains at the industry and aggregate level. We find wide variation in potential productivity gains by industry,

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<sup>2</sup>The Census BTOS Survey estimates between August and November 2024 5.4% of firms had incorporated AI technologies into their production process. This suggests that firm-based measures may miss an important share of genAI adoption. Aaron Levie, the founder and CEO of Box, a cloud storage provider, compares this pattern with past technologies: “It’s almost a one-to-one parallel with the early wave of consumerization of IT. We saw that exact same dynamic in the early 2010s, which was employees of enterprises were pretty frustrated with their traditional document management and collaboration technology, and so they would go out and just go onto a web browser, sign up for a Box account, and then bring it into their organization.” (Levie, 2024)

driven by differences in both the intensity of genAI use and labor cost shares. These factors can offset one another: for example, the information services sector has high rates of genAI use and time savings, but its estimated productivity gain is muted by its low labor cost share. At the aggregate level, we estimate a potential productivity gain of 1.2% based on current genAI use. This is close to Acemoglu (2024), who uses a similar framework to estimate a 0.7% gain based on predicted task exposure rather than actual adoption. Notably, however, Acemoglu (2024) assumed these productivity gains may take a decade to materialize. In contrast, our estimates reflect current adoption and time savings, implying a faster pace of diffusion. We also emphasize that our estimates do not capture potential productivity gains from using genAI to reorganize or completely automate production processes, and therefore likely represent a lower bound over longer horizons (Bresnahan and Trajtenberg, 1995; Brynjolfsson and Hitt, 2000; David, 1990; Jovanovic and Rousseau, 2005).

## 2 Data Sources and Measurement

### 2.1 The Real-Time Population Survey (RPS)

Our main data source is the Real-Time Population Survey (RPS), a national labor market survey of U.S. adults aged 18-64 (for a detailed discussion, see Bick and Blandin 2023). The RPS is fielded online by Qualtrics, a large commercial survey provider, and has collected multiple survey waves each year starting in 2020.

The RPS is designed to mirror the Current Population Survey (CPS) along key dimensions. The RPS matches questions on demographics and labor market outcomes in the basic CPS and CPS Outgoing Rotation Group, using the same word-for-word phrasing when practical and replicating the intricate sequence of questions necessary to elicit labor market outcomes in a manner consistent with the CPS (US Census Bureau, 2015). Replicating key portions of an existing high-quality survey ensures that survey concepts are comparable, which allows researchers to validate RPS outcomes against a widely used benchmark with a larger sample size and to construct sample weights. Bick and Blandin (2023) and Bick et al. (2025a,b) show that the RPS closely aligns with government household surveys on distributions of employment, hours worked, earnings, industry composition, employee tenure, and work from home.

In June 2024, the RPS introduced a novel module measuring genAI use both at work and at home.

### 2.1.1 RPS Sample

Qualtrics panel members are recruited to the panel online and, in our case, can participate in exchange for 30 to 50 percent of the \$5.50 paid per completed survey. The Qualtrics panel includes about 15 million members and is not a random sample of the U.S. population. However, Qualtrics can target survey invitations to specific demographic groups. The RPS sample was designed to be nationally representative across key demographics. We also include measures to filter out low quality responses. Details on sampling and data quality procedures can be found in Appendix [A.1](#).

We fielded the RPS with a pilot genAI module in June 2024 and received 2,551 responses. We then launched the full genAI module with the August and November 2024 waves and received 5,014 and 5,329 responses respectively. All surveys started fielding during the same weeks that the CPS conducted its corresponding surveys. Because the results are very similar across all surveys but our June pilot featured fewer questions, we report results from the pooled August and November surveys. Appendix [B](#) replicates the main figures of the paper separately by survey waves.

Appendix Table [A.1](#) compares the unweighted sample composition between the CPS and RPS along the demographics targeted in the sampling procedure for our main surveys (columns 1 and 2). The bottom panel also compares employment status in the CPS and RPS, which was not targeted in the sampling procedure. The two surveys yield very similar population shares.

### 2.1.2 Sample Weights and Validation

To address remaining discrepancies, we construct sample weights using the raking algorithm of Deming and Stephan ([1940](#)), ensuring that the weighted sample proportions align with the demographic characteristics targeted in the sampling procedure and that the employment shares in the RPS align with the CPS as well. Details on weighting can be found in Appendix [A.2](#).<sup>3</sup>

In Appendix [A.3](#), we show that the distribution of usual weekly earnings, occupation, and industry shares in the RPS, which were not targeted during sampling, closely align with the CPS, in both the raw and weighted data.

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<sup>3</sup>In particular, Appendix [A.2.1](#) discusses a weighting adjustment we applied based on an experimental survey which randomized the sequencing of questions in the genAI module.

## 2.2 Measurement of Generative AI Use

When designing our survey, a key goal was to enable comparisons to the historical adoption of other technologies. To facilitate these comparisons, we use existing technology adoption questions from the CPS Computer and Internet Use supplement (CIU) as a template for our own questions regarding genAI. In the CIU, the leading question about computer adoption at work for employed respondents was:

*Do you [directly] use a computer for your job? (No/Yes)*

A second question asks about computer use at home:

*Do you [directly] use a computer at home? (No/Yes)*

These questions were asked in 1984, 1989, 1993, 1997, 2001, and 2003, with the word “directly” being omitted in the last two years.<sup>4</sup>

The RPS genAI module begins with a definition of genAI:

*Generative AI is a type of artificial intelligence that creates text, images, audio, or video in response to prompts. Some examples of Generative AI include ChatGPT, Gemini, and Midjourney.*

We included examples of popular genAI products because we thought some respondents may be more familiar with those product names than with the broader concept of genAI. After defining genAI, the module asks respondents whether they had heard of the concept prior to the survey. Overall, 88.4% reported having heard of genAI; the remaining 11.6% who answered “No” skip the remainder of the module. Respondents who answer “Yes” continue with the survey.

For employed respondents, the next question asks about genAI use at work:

*Do you use Generative AI for your job? (No/Yes)*

This question is designed to mirror the analogous computer use question from the CIU, discussed above. We opted for the version asked from 2001 onwards that omitted the word “directly.”

Another question later in the survey asks about genAI usage outside work:

*Do you use Generative AI [outside your job]? (No/Yes)*

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<sup>4</sup>Starting in 2001, the CIU also asked about internet usage.

Non-employed respondents are not shown the term in brackets. While the CIU asks about computer use “at home”, we use slightly different language to account for respondents who use genAI on mobile devices outside of their home.

The survey asks several follow-up questions of genAI users regarding which products they used, how intensively they used it, and how much time they saved, always distinguishing between work and non-work use.

An important caveat regarding our measurement approach is that it will only capture genAI use that respondents are aware of. For example, while typing prompts into ChatGPT is an obvious case of using genAI, automatic AI-assisted writing suggestions are a more subtle case that respondents may not consider. Because our estimates may not detect some passive or embedded uses of the technology, we interpret our estimates a lower bound on the extent of genAI adoption.

### 3 How Prevalent is Generative AI Use?

Figure 1a displays the share of August and November 2024 RPS respondents who report using genAI. The samples for the “Overall” and “Outside of Work” bars are all respondents, while the sample for the “For Work” bar is employed respondents. The overall estimates reflect the share who use genAI either for work or outside of work.

The first bar shows that 45.5% of respondents report using genAI either at work or at home: 10.4% used it every day, 26.7% used it some but not all days, and 8.3% did not use it in the last week. The second bar shows that 32.1% of workers used genAI at work: 10.5% used it every work day and 16.7% on some but not all workdays, bringing total weekly usage to 27.2%. Outside of work, usage is higher overall (37.7%) but less intensive, with only 5.6% of all respondents using it every day last week.

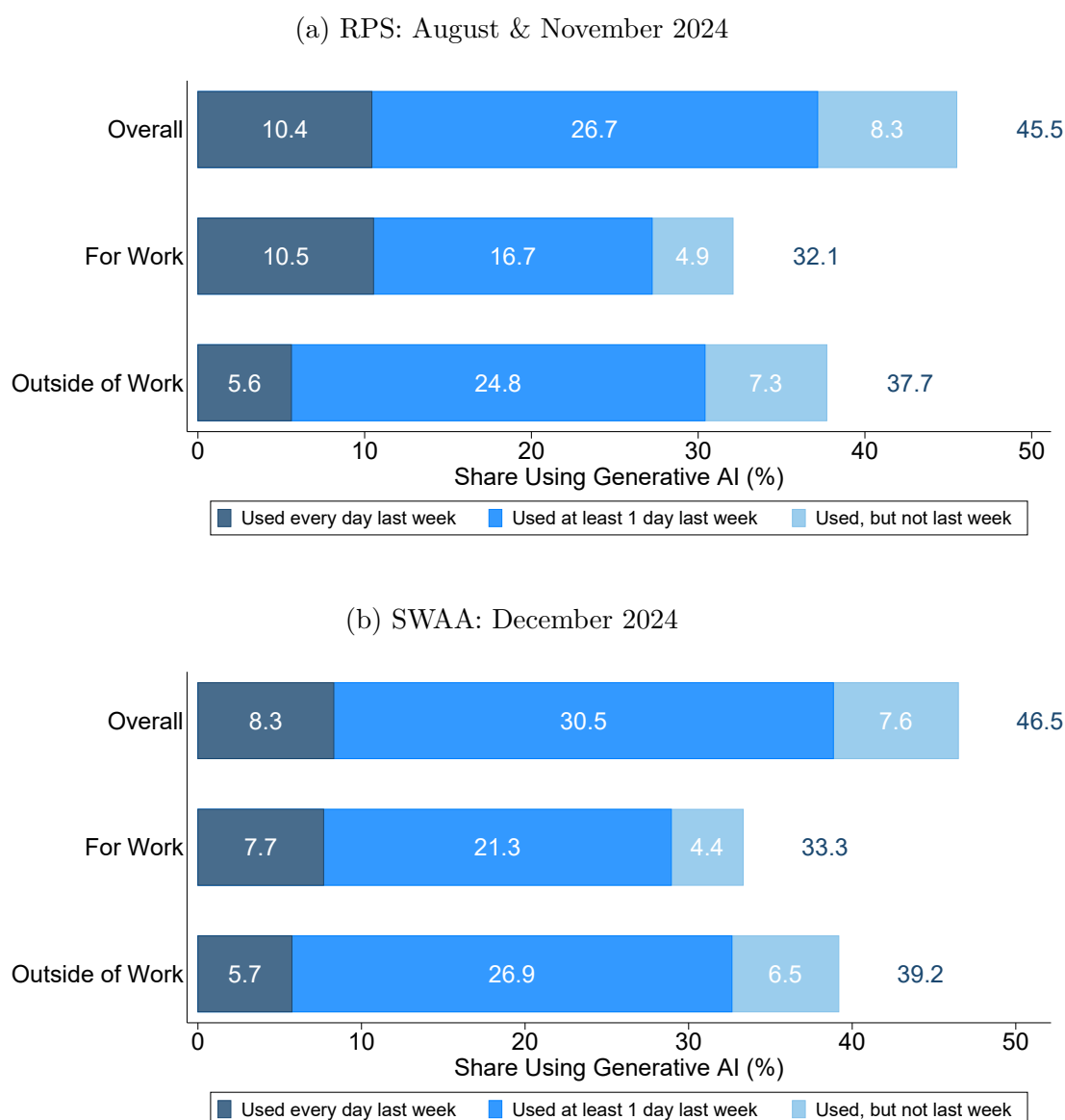
#### 3.1 How Do Our Results Compare to Other Estimates?

The RPS is administered by Qualtrics. To provide a second source of data, in December 2024 we fielded our genAI module through the Survey of Working Arrangements and Attitudes (SWAA) by Barrero et al. (2021).<sup>5</sup> The SWAA uses a different survey provider (IncQuery) but targets a similar sample intended to be representative of the U.S. population ages 20-64. Figure 1b displays analogous results from the SWAA. 46.5% of SWAA respondents reported using genAI overall, versus 45.5% of RPS respondents. We find similar estimates for work (33.3% SWAA,

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<sup>5</sup>See Appendix C for more details on the SWAA, how we construct weights to maximize comparability between the SWAA and RPS, and some validation exercises.

Figure 1: Share of Working Age Adults Using Generative AI



*Notes:* The figure shows the share of respondents who use genAI for work, outside of work, and overall (either for work or outside of work). Intensity of use is broken down into every day last week (dark blue), at least one day but not every day last week (medium blue), and not last week (light blue). Data source for panel (a) is the August and November 2024 waves of the RPS, ages 18-64. The “For Work” sample is employed individuals ( $N = 6935$ ); the other bars include all respondents ( $N = 9726$ ). Data source for panel (b) is the December 2024 wave of the SWAA, ages 20-64. The “For Work” sample is employed individuals ( $N = 3516$ ); the other bars include all respondents ( $N = 4698$ ).

32.1% RPS) and outside work (39.2% SWAA, 37.7% RPS). Appendix C.3 includes additional comparisons between the SWAA and RPS. Because the two surveys yield similar estimates but the SWAA only includes a subset of the genAI questions in the RPS, the remainder of the main text reports results only from the RPS.

We presented genAI users a list of common products (and a write-in option) and asked them to select (all) the products they use. This allows us to compare ChatGPT use in the RPS with other surveys. The three most used genAI products were ChatGPT (32% of all respondents), Gemini (19%), and embedded tools like Microsoft Copilot (15%). Appendix Figure C.2a shows the full distribution of product usage. In April 2024, Fletcher and Nielsen (2024) conducted an online survey of ChatGPT adoption using a different survey provider (YouGov). They found that 18% of U.S. adults used ChatGPT at least weekly, compared to 22% who used ChatGPT in the previous week in our Fall 2024 surveys.<sup>6</sup>

A potential concern with online-based surveys is that they may suffer from selection based on unobservable characteristics that are correlated with genAI use. For example, online survey respondents may be more comfortable using technology and therefore be more likely to use genAI than the overall U.S. population. From this perspective, a valuable point of comparison is a survey conducted by the Pew Research Center (McClain, 2024). That survey sampled respondents using a nationally representative sample of U.S. Postal Services addresses. Importantly, the survey included respondents without internet access. (Internet access per se is unlikely to substantially impact our results: in the 2022 ACS, 96.8% of individuals aged 18-64 reported having access to the internet.) They find that in February 2024, 27% of U.S. adults age 18-64 reported ever having used ChatGPT, compared to 32% of individuals who report using ChatGPT in our survey in August and November 2024.

Finally, we can also compare genAI adoption by workers to adoption by firms. The monthly Census BTOS Survey asks firms: “In the last two weeks, did this business use Artificial Intelligence (AI) in producing goods or services? (Examples of AI: machine learning, natural language processing, virtual agents, voice recognition, etc.)” (Bonney et al., 2024). Between August and November 2024, 5.3% of firms report using AI, compared with 32.1% of workers who report using genAI in the RPS. While differences in question wording may explain part of this gap, they are unlikely to account for all of it.<sup>7</sup> We conclude that, so far, genAI adoption by workers is outpacing firm-wide adoption, implying that firm-based measures may miss an important share of genAI’s impact (Babina et al., 2024).

### 3.2 How Does Early Generative AI Adoption Compare to Other Technologies?

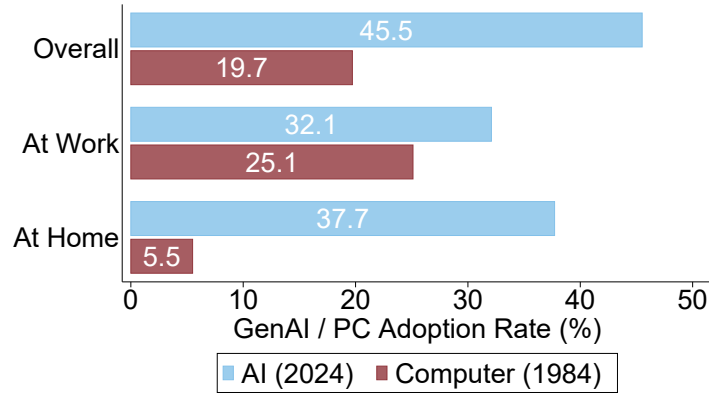
Figure 2 compares the pace of genAI adoption with PCs two to three years since the release of the first-mass market product for each technology. (As described in Section 2.2, adoption was

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<sup>6</sup>Humlum and Vestergaard (2024) survey a representative sample of workers in eleven occupations in Denmark about their usage of ChatGPT at work. We find similar patterns by age, education and gender in the occupations covered by both surveys.

<sup>7</sup>For example, some respondents in the BTOS survey may not interpret using a genAI tool in a company’s human resource department as being used “in producing goods or services.”

Figure 2: Computer and Generative AI Adoption 2-3 Years After First Mass-Market Product



*Notes:* The figure displays usage rates for computers (PCs) and genAI in 1984 and 2024, respectively. The IBM PC was released in 1981, which we refer to as the first mass-market computer, implying that 1984 is three years after the first mass-market computer. ChatGPT was released in November 2022, which we refer to as the first mass-market genAI product, implying the fall of 2024 is two years after the first mass-market product for that technology. The data source for genAI is the August and November 2024 wave of the RPS (blue bars). The data source for computers is the 1984 Computer and Internet Use Supplement of the CPS (red bars). The sample for the Overall and At Home bars in the RPS and CPS is all individuals ages 18-64 (RPS,  $N = 9726$ ; CPS,  $N = 61,708$ ). The sample for the For Work bars is employed individuals ages 18-64 (RPS,  $N = 6935$ ; CPS,  $N = 89,271$ ).

measured using very similar questions for both technologies.) The first mass-market computer was the IBM PC, which was released in August 1981. The first mass-market genAI product was ChatGPT, which was released in November 2022.<sup>8</sup> For the PC, we measure adoption in the CPS CIU for the first time in 1984, i.e. three years after the first mass-market computer. The RPS provides adoption estimates for 2024, two years after the first mass-market genAI product.

The top pair of bars in Figure 2 refers to overall adoption for the two technologies two to three years after the introduction of the first mass-market product. The bar for the RPS repeats the 45% overall adoption rate for genAI reported in Figure 1a. Overall PC adoption is roughly 20% in 1984, implying that genAI has been adopted at a faster pace than the PC. As another more recent comparison, in 1995 the internet was opened to commercial traffic after the decommissioning of NSFNet (Leiner et al., 2009).<sup>9</sup> Two years later, 21% of US adults reported access to the internet according to the International Telecommunication Union, which compiles subscriber data from national regulatory bodies and service providers (Pena-Lopez et al., 2009). This is a similar pace of overall adoption to PCs, and substantially slower than the adoption of genAI.

<sup>8</sup>Three notable computer products had already been released in 1977 (the Apple II, the Commodore PET, and the TRS-80). Choosing earlier introduction dates would imply slower adoption rates for a given technology. The 1981 IBM PC was the first computer to sell over one million units (Abbate, 1999). We chose these dates based in part on ChatGPT’s answers to questions about the year of the first mass market product for each technology. The exact prompts and answers are available upon request.

<sup>9</sup>1995 was also the year of Netscape’s initial public offering and the year that AOL 3.0 was released.

A key question is whether genAI’s faster adoption was driven by work or non-work use.<sup>10</sup> The faster overall pace of genAI adoption compared with PCs is driven by non-work adoption. The middle pair of bars in Figure 2 plots work adoption of genAI and PCs. We find similar work adoption rates for genAI after two years (32%) and PCs after three years (25%). By contrast, the bottom pair of bars in Figure 2 reveals much higher non-work adoption rates for genAI (38%) compared with PCs (5%).

The computer and related information technologies induced far-reaching changes in firm organization and management, which led to substantial productivity gains (Brynjolfsson and Hitt, 1996). The similar pace of genAI adoption by workers thus far suggests that this new technology could also be transformative for firms. At the same time, the choice of whether to adopt a technology depends on both its benefits and costs. PCs required purchasing expensive immobile hardware and nontrivial training. In contrast, many genAI products are free or inexpensive and user-friendly. On the one hand, this lower adoption cost likely contributes to genAI’s rapid uptake. On the other hand, this means that similar adoption rates for PCs and genAI at work do not necessarily imply equivalent benefits.

## 4 Which Workers Use Generative AI?

In this section, we document that early genAI adoption has not been uniform throughout the workforce. Understanding which workers are adopting genAI at higher rates provides several important insights. First, because many workers are adopting genAI ahead of firm-level adoption, variation in genAI use can help managers allocate support, training, and integration efforts. Second, we show that employer encouragement of genAI use is highly predictive of adoption, suggesting that firms and managers can influence the nature and extent of genAI use in their organization. Third, adoption patterns shed light on how genAI maps onto specific tasks and roles within the firm, which is relevant for decisions regarding job redesign and firm reorganization.

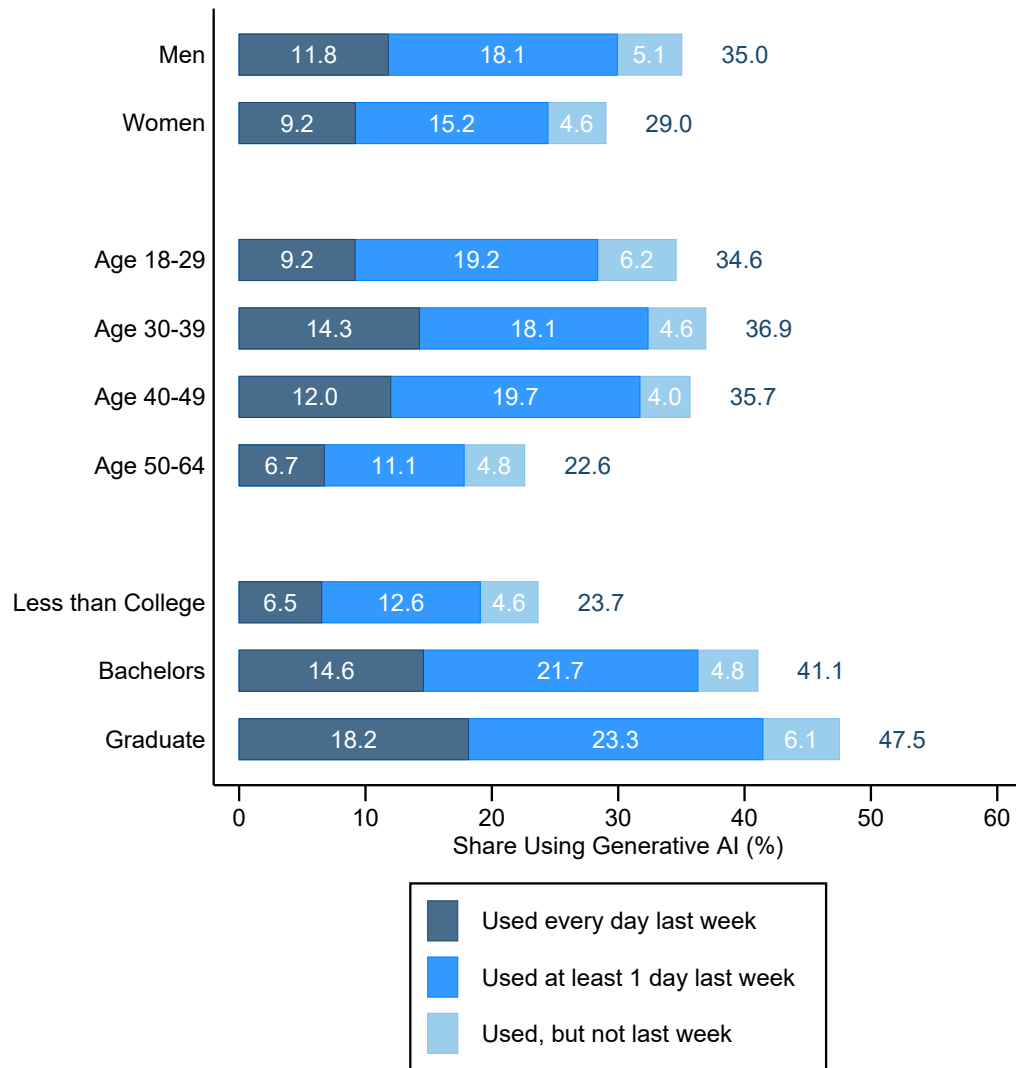
Figure 3 documents demographic heterogeneity in work adoption of genAI. The top panel shows genAI adoption is 6 percentage points higher for men. The middle panel show genAI adoption by age. Workers aged 18-49 have similar adoption rates, while adoption is lower for workers aged 50-64. The bottom panel shows that adoption rates are increasing in educational attainment. In particular, workers with a Bachelor’s degree or more use genAI at roughly twice the rate of non-college educated workers. Differences between demographic groups are primarily driven by users who use genAI every workday or at least once in the previous week.

Figure 4 plots the kernel density of genAI adoption by weekly earnings percentile. Adoption

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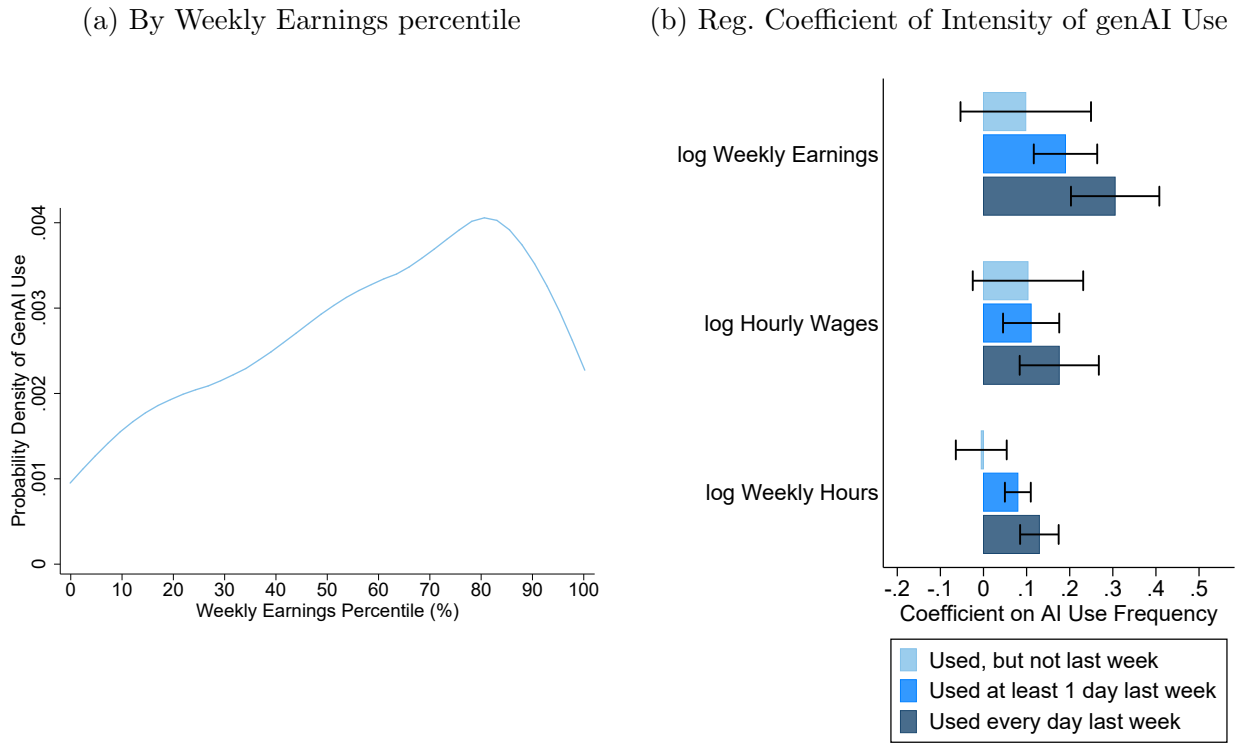
<sup>10</sup>Unfortunately, internet adoption data do not allow us to distinguish work from non-work adoption.

Figure 3: Demographic Heterogeneity in Work Adoption of Generative AI



*Notes:* The figure shows adoption rates at work for genAI by sex, age, and education. The data source is the August and November 2024 waves of the RPS. The sample is employed individuals ages 18-64 (RPS,  $N = 6935$ ).

Figure 4: Earnings and Generative AI Adoption



*Notes:* Panel (a) shows adoption rates at work for genAI by weekly earnings percentile (kernel density). Panel (b) shows the coefficients on genAI intensity-of-use (zero days last week, some but not all workdays last week, and every workday last week) in a regression of (i) log weekly earnings, (ii) log hourly wages, and (iii) log weekly hours worked. For all regressions, the additional control variable used are indicators for: female, age 18-29, age 40-49, age 50-64, a bachelor's degree, a graduate degree, Black, Hispanic, occupation group, and industry. The data source for genAI is the August and November 2024 waves of the RPS. The sample is employed individuals ages 18-64 whose implied hourly wage is at least the federal minimum wage and whose weekly earnings is below \$10,000 (RPS,  $N = 6063$ ).

rates increase steadily with income until roughly the 80th percentile, and then decline. To better understand the relationship between technology use and earnings, we regress earnings, wages, and hours on indicators of the frequency of genAI use while also controlling for occupation, industry, education, and demographic characteristics.<sup>11</sup> The results are displayed in Figure 4b. On average, genAI use is associated with a weekly earnings premium of 20 log points, a wage premium of 15 log points, and 5 log points for weekly hours. However, these average effects mask important heterogeneity. The coefficients on earnings, wages, and hours are all increasing in genAI use intensity, and infrequent users show no statistical difference between non-users.

Higher wages by (more intensive) users could reflect several factors. First, within an occupation and industry jobs that benefit more from genAI may be associated with higher wages (Deming and Kahn, 2018). Second, more productive workers may adopt new technologies more

<sup>11</sup>The control variables used in these regressions are indicators for: female, age 18-29, age 40-49, age 50-54, a bachelor's degree, a graduate degree, Black, Hispanic, occupation group, and industry.

rapidly (Nelson and Phelps, 1966). Third, genAI may directly increase workers’ productivity and wages. Our data do not allow us to distinguish between these explanations.

#### 4.1 Tasks, Predicted Exposure, and Generative AI Adoption

Why do some groups of workers adopt genAI at higher rates than others? One reason could be that workers differ in their job tasks or responsibilities, and genAI is more helpful in assisting some tasks than others.

Consistent with this explanation, we first document that genAI adoption differs greatly between occupations; see Figure 5. The three occupations with the highest genAI adoption are computer and mathematical, management, and business and finance occupations (54%, 52%, and 48%, respectively) and the bottom three are blue collar, office and administration, and personal services occupations (26%, 23%, and 16%, respectively). At the same time, genAI adoption exceeds 20% in all but one occupation groups, suggesting that the technology is useful for a broad range of occupations. As with demographic differences, differences between occupations are driven by the rates of daily and weekly genAI users.

Do different adoption rates between occupations reflect differences in tasks or responsibilities? One approach to assess this is to construct task-based-exposure measures across occupations (Eloundou et al., 2024; Felten et al., 2021; Webb, 2020). For example, Eloundou et al. (2024) constructs a rubric that assigns each task in the O\*NET database a zero or one corresponding to whether LLMs (a popular class of genAI) could feasibly produce large productivity gains for that task. They then construct a predicted exposure score for each occupation by aggregating task-level scores for each occupation. Using RPS data, we can ask whether these predictions are correlated with adoption. Our study is the first to evaluate these task-based “exposure” predictions using actual genAI adoption rates.<sup>12</sup>

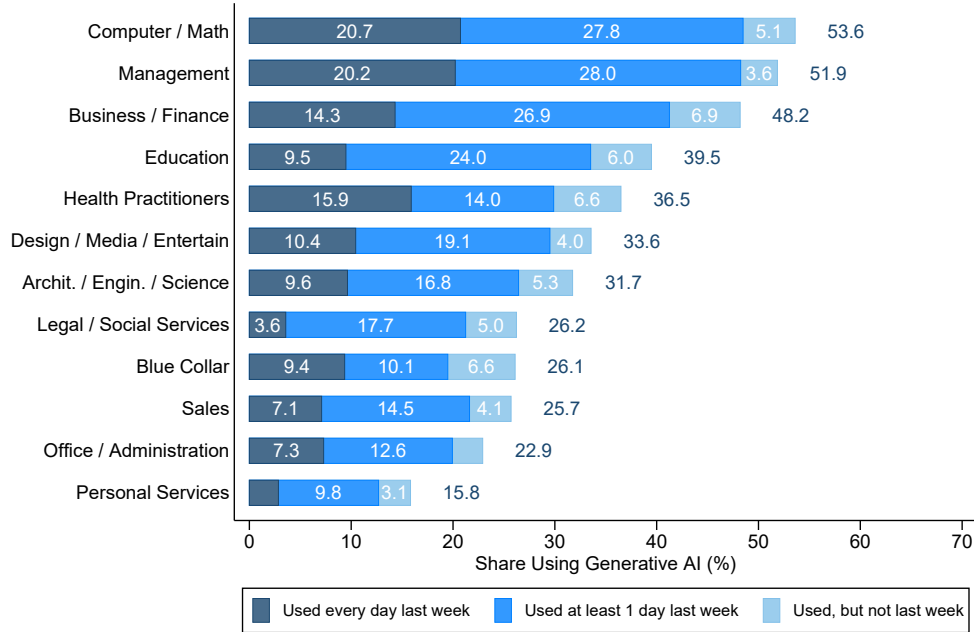
Figure 6a plots an occupation group’s predicted genAI exposure from Eloundou et al. (2024) against actual genAI adoption in the 2024 RPS. We find that predicted genAI exposure is highly correlated with actual genAI adoption ( $\rho = 0.65$ ). In Figure 6b, we find a similar correlation ( $\rho = 0.71$ ) using the predicted exposure score from Felten et al. (2021). For example, personal service and blue-collar occupations have relatively low predicted exposure and also relatively low adoption, while computer and mathematical occupations have both high predicted exposure and high adoption.

The high correlation between predicted exposure and actual adoption provides partial validation of task-based exposure predictions, supporting their utility for researchers. At the same

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<sup>12</sup>Handa et al. (2025) assign tasks to conversations by Claude.ai Free and Pro users and show that task usage rates are correlated with the predictions by Eloundou et al. (2024).

Figure 5: Generative AI Work Adoption By Occupation



*Notes:* The figure shows adoption rates at work for genAI by occupation. The data source is the August and November 2024 waves of the RPS (blue bars). The sample is employed individuals age 18-64 (RPS,  $N = 6883$ ). We exclude workers in “agriculture / forestry / fishing” due to a very low sample size. For details on occupation categories, see Appendix A.4.

time, we also note particular occupations whose adoption rates deviate from what we would expect based on their exposure score. In particular, office and administrative occupations and managers are predicted to be similarly exposed to genAI, but actual adoption by managers is more than double that of office and administrative workers. One potential reason for these discrepancies could be that existing task-based predictions do not perfectly identify which occupations would benefit from genAI.

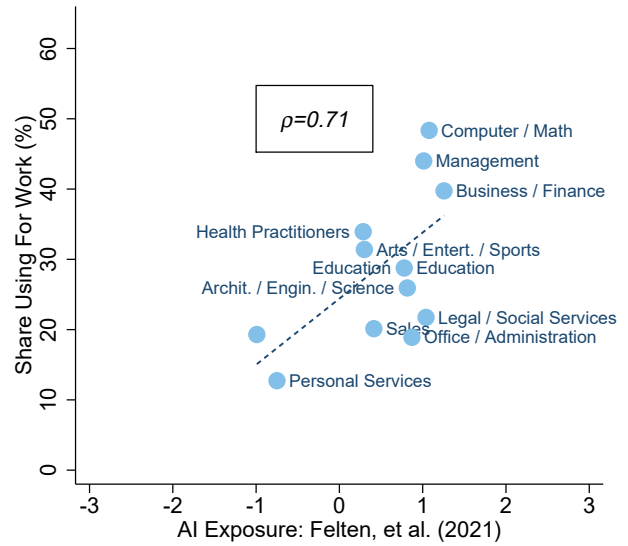
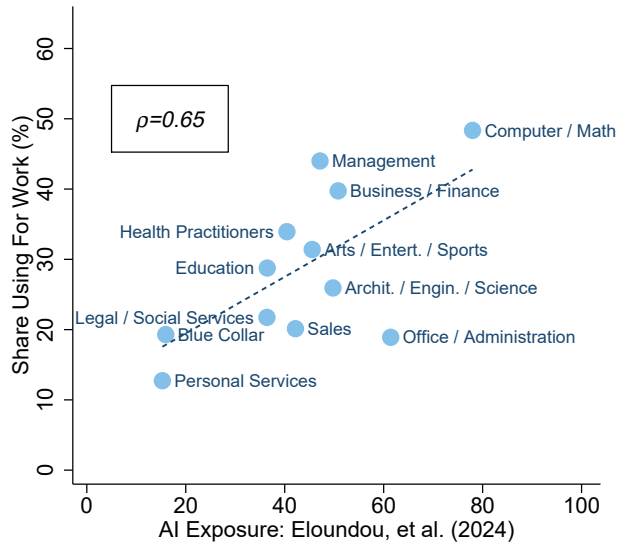
To investigate this possibility, we leverage additional information about workers’ adoption decisions from the RPS. Among users who did not adopt genAI, we provided a list of potential reasons why they did not adopt. We then asked respondents to select the reasons that were relevant for their decision and to rank them in order of importance. We define the share of an occupation who does not use genAI because it is not useful for their job as the share of all workers in that occupation who ranked “GenAI cannot help me with my job” in their top two reasons for not using genAI. (For the full list of potential reasons, see Appendix A.5.)

Figure 6c plots genAI adoption against the share who reported that it was not useful. We find that occupational adoption is highly correlated ( $-0.88$ ) with the share who reported genAI was not useful. For example, only 7% of workers in computer / math occupations reported that genAI was not useful for their job, compared with 30% of personal service workers.

Figure 6: Accounting for Variation in Generative AI Adoption At Work By Occupation

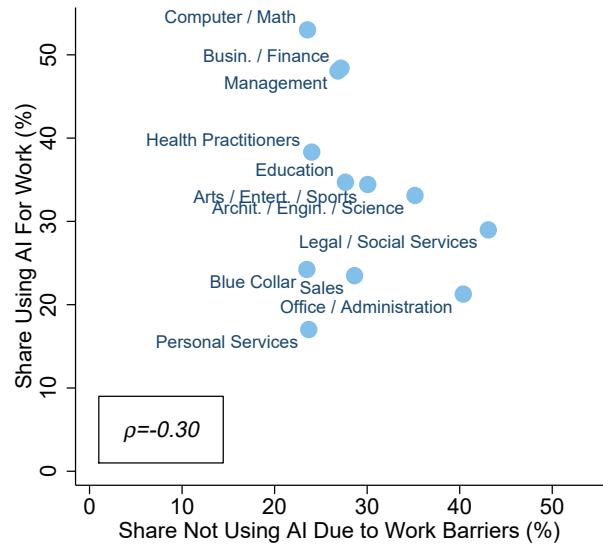
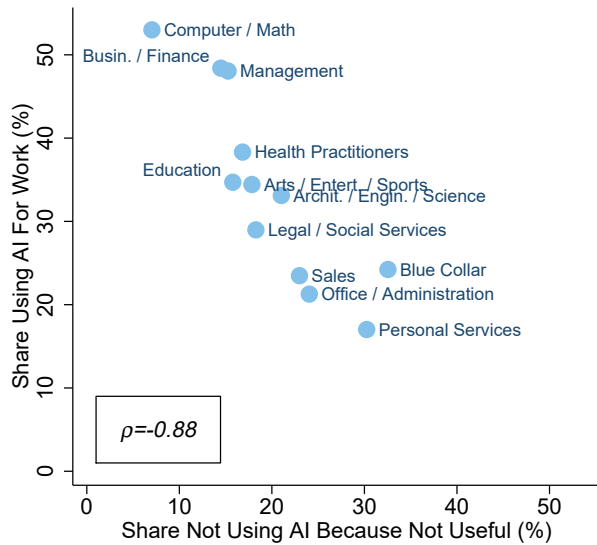
(a) Predicted Exposure: Eloundou et al. (2024)

(b) Predicted Exposure: Felten et al. (2021)



(c) Share Stating AI Not Useful

(d) Share Stating AI Barriers at Work



*Notes:* Panel (a) scatters occupational genAI adoption rates at work against predicted genAI exposure (ChatGPT- $\beta$ ) from Eloundou et al. (2024). Panel (b) scatters occupational genAI adoption rates at work against predicted genAI exposure from Felten et al. (2021). Panel (c) scatters occupational genAI adoption rates at work against the share of all workers in that occupation who reported that they did not use genAI because it was not useful for their job. Panel (d) scatters occupational genAI adoption rates at work against the share of all workers in that occupation who reported that they did not use genAI due to barriers at work. The data source for genAI use and reasons for non-use is the November 2024 RPS. The sample for the RPS is employed individuals age 18-64 ( $N = 3671$ ). We exclude workers in “agriculture / forestry / fishing” due to a very low sample size.

These results are consistent with the notion that existing task-based exposure predictions may not perfectly identify which occupations would benefit from genAI. For example, while managers have similar predicted exposure than administrative occupations, many more administrative workers report that genAI is not useful for their job. Collectively, our takeaway is that existing task-based exposure measures are informative but incomplete: while they do capture broad patterns in genAI adoption, more work is needed to accurately identify which jobs would benefit from genAI.

Another potential reason that task-based exposure predictions may not perfectly predict adoption is that some jobs may face various barriers to adoption. In our question about why workers did not adopt genAI, a second set of explanations relates to adoption barriers: “My employer does not allow me to use genAI; I am unsure whether my employer allows me to use genAI; I feel pressure from coworkers to not use genAI; I have privacy or security concerns; I am unsure how to use genAI for my job.” We define the share of an occupation who does not use genAI because of adoption barriers as the share of all workers in that occupation who ranked at least one of these options in their top two reasons for not using genAI.

Figure 6d plots genAI adoption against the share who reported that they faced adoption barriers. This correlation (-0.30) is lower than in the previous figure, indicating that variation in adoption barriers only accounts for a small share of the overall variation in adoption. At the same time, two occupations stand out in terms of a high share of workers facing adoption barriers: legal / social services and office / administrative workers. Workers in these occupations may be especially likely to deal with sensitive information that makes employers reluctant to allow genAI use, or leads workers to self-regulate in the absence of formal policies. These findings suggest that managers can play a role in genAI adoption by setting clear usage guidelines and helping employees navigate privacy and security issues.

In addition to minimizing barriers to genAI adoption, we also find evidence that firms and managers can increase genAI adoption through active encouragement. The RPS asked workers “Does your employer encourage you to use Generative AI for your job?”. Employer encouragement is highly correlated with adoption: 86% of respondents whose employer encourages genAI use report adopting it, compared to just 13% among those whose employer does not.

We conclude this section by jointly analyzing the role of demographics, occupation, and employer encouragement. Table 1 presents results from a linear regression of an indicator for genAI use last week on these variables. Demographic controls (sex, age, and education) alone account for 8% of the variation in adoption. Adding controls for occupation group and predicted exposure by occupation (Eloundou et al., 2024) increases this share to 12%. Occupation controls decrease the magnitude of most demographic coefficients, and in particular reduce the coefficients on Bachelor’s Degree and Graduate Degree by roughly one third. Finally,

Table 1: Predictors of Generative AI Use at Work Last Week

|                                      | (1)                  | (2)                  | (3)                  |
|--------------------------------------|----------------------|----------------------|----------------------|
| Constant                             | 0.266***<br>(0.018)  | 0.213***<br>(0.052)  | 0.066<br>(0.042)     |
| Female                               | -0.070***<br>(0.014) | -0.048***<br>(0.015) | -0.018*<br>(0.011)   |
| Age: 18-29                           | -0.016<br>(0.022)    | -0.014<br>(0.021)    | -0.028*<br>(0.016)   |
| Age: 40-49                           | -0.015<br>(0.021)    | -0.028<br>(0.021)    | 0.009<br>(0.016)     |
| Age: 50-64                           | -0.152***<br>(0.019) | -0.157***<br>(0.019) | -0.059***<br>(0.014) |
| Education: Bachelor's Degree         | 0.189***<br>(0.017)  | 0.123***<br>(0.018)  | 0.032**<br>(0.014)   |
| Education: Graduate Degree           | 0.243***<br>(0.022)  | 0.157***<br>(0.024)  | 0.034*<br>(0.018)    |
| Predicted Exposure (Eloundou et al.) | —                    | 0.206***<br>(0.065)  | 0.092*<br>(0.051)    |
| Employer Encourages AI Use           | —                    | —                    | 0.652***<br>(0.016)  |
| Occupation Controls                  |                      | Y                    | Y                    |
| R <sup>2</sup> -adj                  | 0.07                 | 0.12                 | 0.50                 |
| N                                    | 6883                 | 6883                 | 6883                 |

*Notes:* Dependent variable is an indicator for whether the respondent used genAI at work in the previous week. Standard errors are in parentheses. \*, \*\*, \*\*\* indicate significance at the 10, 5, and 1 percent levels, respectively. Data source is the August and November 2024 waves of the RPS, ages 18-64. The sample for this table is employed individuals. We exclude workers in “agriculture / forestry / fishing” due to a very low sample size.

adding an indicator for employer encouragement dramatically raises the share of variation explained to 50%. The coefficient on employer encouragement indicates that, holding other variables constant, a worker is 65 percentage points more likely to use genAI for work if their employer encourages them to do so. Controlling for employer encouragement also reduces the magnitude of coefficients on demographic and occupation controls relative to the previous specification. We caution that the estimated effect of employer encouragement is not necessarily causal as it may be highly correlated with a worker’s roles and responsibilities (though this is partially addressed through occupation controls).

The findings in this section support two conclusions. First, job tasks and responsibilities appear to be a primary determinant of genAI adoption patterns. Second, workplace barriers and encouragement also play an important role in a worker’s adoption decision. Hence, when job tasks are amenable to genAI adoption, firms can play an active role by promoting usage, offering training, and removing organizational barriers to adoption.

Figure 7: In Which Specific Work Tasks Is Generative AI Most Useful?



*Notes:* The figure shows which work tasks genAI users report that genAI is most helpful in completing. Employed respondents who used genAI were first provided with a list of tasks and asked to select those that they had used genAI to help with last week. Respondents were then asked to rank these selected tasks according to how helpful genAI was in completing the task. The figure reports the share of genAI users who ranked a particular task either #1 (AI was most helpful in this task) or #2. The bars do not have a natural sum because some respondents selected fewer than two tasks. Data source is employed respondents ages 18-64 who used genAI for work and selected at least one task in the August and November 2024 waves of the RPS.

## 4.2 Generative AI Use Across Specific Tasks

Given the high correlation between job tasks and adoption, a natural follow up question is to ask which tasks genAI helps workers complete. We present genAI users with the list of tasks in Figure 7 and ask them to select any for which they had used genAI in the last week. (They were also allowed to write in other tasks.) Respondents were then asked to rank these tasks according to how helpful genAI was in completing them.

Figure 7 reports the share of respondents who ranked each task in the top two in terms of importance. The highest ranked tasks at work were writing (45.1%), administrative tasks (29.9%), and interpreting/translating/summarizing text or data (25.9%). These findings mirror recent experimental evidence on how genAI reshapes work behavior. Dillon et al. (2025) show that workers with access to integrated generative AI tools (Microsoft 365 Copilot) spent less time on email and completed written documents more quickly. At the same time, while genAI seems most useful for creating, absorbing, and organizing written information, eight of the ten tasks in our list were ranked by at least 10% of users, indicating a variety of use cases for genAI.

## 5 How Large Are Potential Productivity Gains from GenAI?

New technologies raise productivity by enabling complementary innovations in processes, skills, and organizational practices. These innovations typically evolve gradually over time (Brynjolf-

sson et al., 2021). To gain early insights into potential productivity gains from genAI, this section documents worker-reported time savings and shows how this data can inform estimates of aggregate and industry/firm-level productivity gains.

## 5.1 How Much Do Workers Use Generative AI?

We begin by documenting sizable variation in the intensity of genAI use among workers. This is important because, as we show in the next section, workers who use genAI more intensively tend to experience greater time savings.

Figure 8a reports how much time genAI users spent using the technology in a given day. Specifically, we ask “*Please think back to the days LAST WEEK on which you used Generative AI at work. On average, how much time did you spend actively using Generative AI at work?*” Respondents could select from three options: 15 minutes or less per day, between 15 minutes and one hour per day, or more than an hour per day.<sup>13</sup>

Figure 8a shows that 36.9% of genAI users report using genAI for an hour or more per day at work, 42.8% used it for between 15 and 60 minutes per day, and 20.3% used it for less than 15 minutes per day. We also show these estimates separately by workers’ daily frequency of use. Frequency and intensity of genAI use are positively correlated. 65.7% of daily genAI users report using it an hour or more each day, compared to only 6.4% of those who used it in the last month but not the last week. Overall, our results indicate wide variation in the frequency and intensity of genAI use at work.

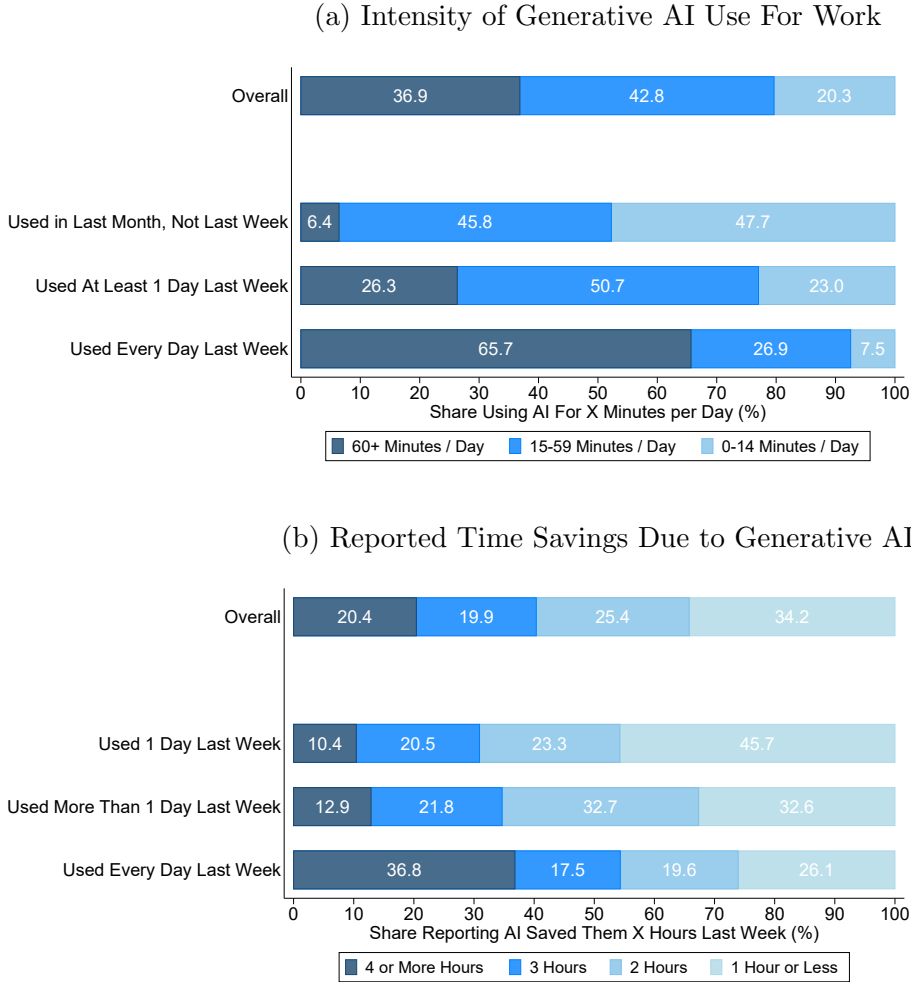
We estimate bounds on the share of total work hours assisted by genAI by combining data on usage intensity with data on days and hours worked in the previous week. (For this calculation, we restrict attention to our November 2024 wave, which contains more detailed intensity categories; see Appendix Figure B.2). For example, consider a worker who reports (i) working 40 hours last week over 5 days, (ii) using genAI on multiple but not all workdays last week, and (iii) using genAI for between 15 and 59 minutes per day on days that they used it. The lower bound implies the respondent used genAI on two days for 15 minutes each day ( $15/59 \cdot 2/40 = 1.3\%$  of their weekly hours); the upper bound implies using genAI on four workdays for 1 hour each day ( $1 \cdot 4/40 = 10\%$  of their weekly hours).

We first aggregate these estimates across the 26.4% of workers in the November 2024 wave who report using genAI in the past week. Among workers who use genAI, between 4.9 and 19.2% of all work hours were assisted by genAI. Next, we include the 73.6% of non-genAI-workers in our calculation, who spend no time using genAI by construction. Among all workers, between

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<sup>13</sup>Respondents who did not use genAI last week but did use it in the last four weeks were asked about their average usage on the days they used genAI over the last four weeks.

Figure 8: GenAI Intensity and Reported Time Savings



*Notes:* Panel (a) shows the distribution of daily time spent actively using genAI for work, among genAI users. The “Overall” bar reflects the distribution among all workers who use genAI. Other bars break down daily time use by daily frequency of use. Data source is the November 2024 waves of the RPS, ages 18-64. The sample is employed respondents who use genAI for work ( $N = 1000$ ). Panel (b) shows the distribution of work time saved in the previous week due to genAI, among individuals who use genAI for work. The “Overall” bar reflects the distribution among all workers who use genAI. Other bars break down time savings by daily frequency of use. Data source is the November 2024 wave of the RPS, ages 18-64 (the August 2024 wave did not ask about time savings). The sample is employed respondents who used genAI for work in the last week ( $N = 925$ ).

1.6% and 6.6% of total work hours were assisted by genAI.

## 5.2 Reported Time Savings Due to Generative AI

In the November 2024 wave, we asked workers who used genAI in the previous week to estimate how much time genAI saved them in the past week:

*You indicated that LAST WEEK you worked   X   hours and that you used Generative AI for your job. Now, imagine that LAST WEEK you did not have access to Generative AI. How many additional hours of work would you have needed to complete the same amount of work? [For X we fill in the respondent’s reported hours worked last week.]*

The answer options were: Less than 1 hour, 1 hour, 2 hours, 3 hours, 4 hours, More than 4 hours. For respondents who selected “less than 1 hour”, we assume that genAI saved zero hours of work. For respondents who selected “more than 4 hours,” we assume that genAI saved 4 hours of work. These assumptions are conservative in the sense that they reflect a lower bound of time savings.

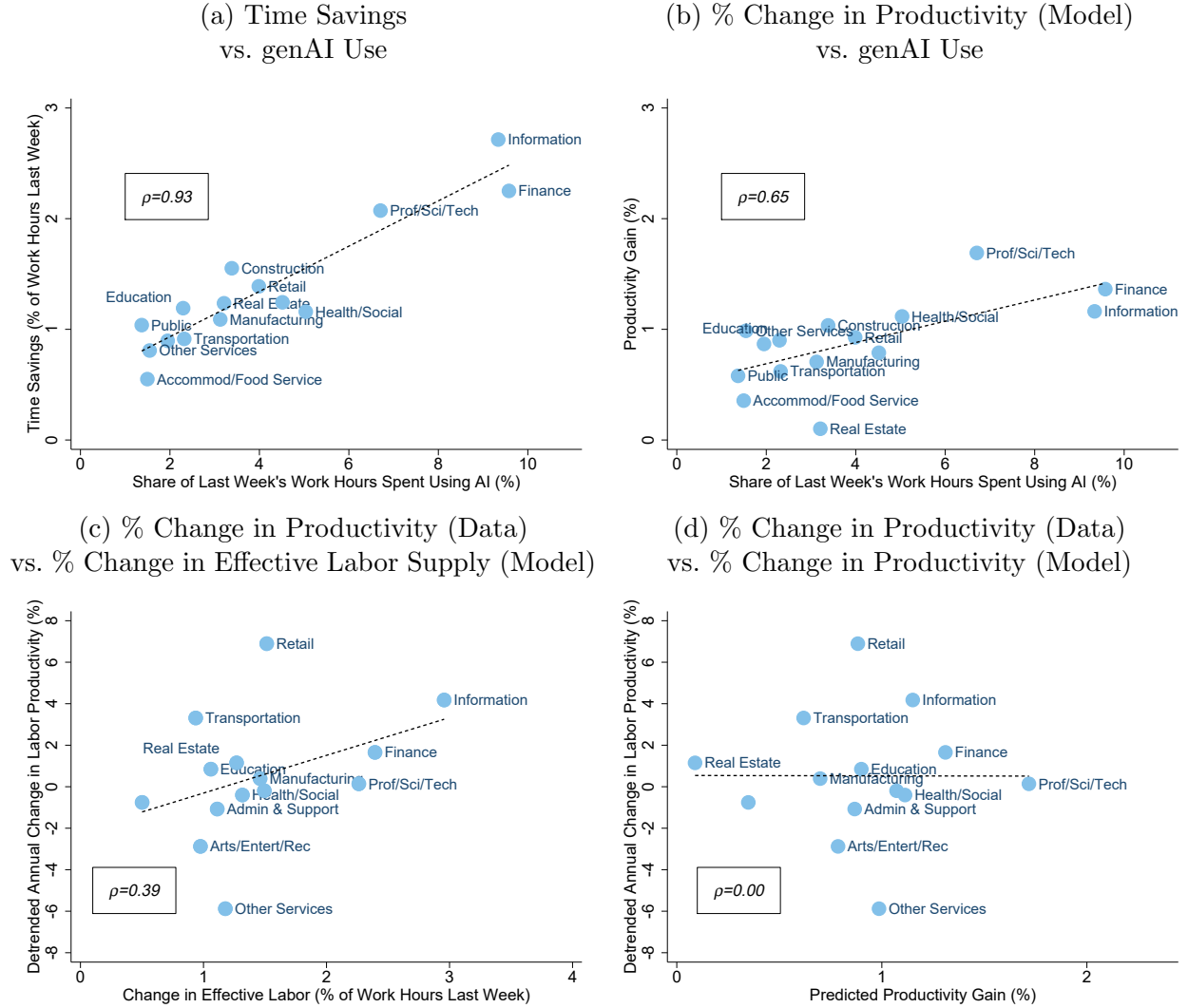
Figure 9 reports the results: 34.2% said genAI saved them an hour or less last week, 25.4% reported two hours, 19.9% reported three hours, and 20.4% reported four hours or more. More frequent users report higher average time savings. Among workers who used genAI every day last week, 36.8% reported saving four hours or more, compared to 10.4% of respondents who used it one day last week.

For each genAI user, we compute the share of working hours saved as the ratio of time saved last week to hours worked last week. We find an average time savings of 5.2% of work hours by genAI users in the November 2024 wave, or about 2.1 hours for a 40-weekly-hour worker. When we include all workers, including non-users, on average genAI saved workers saved 1.4% of total hours.

## 5.3 Potential Productivity Gains from Generative AI By Industry

We now present a standard production model to demonstrate how our data on genAI use and time savings can be used to estimate productivity gains on the industry or aggregate level. Detailed derivations can be found in Appendix D.1.

Figure 9: Generative AI Use, Time Savings, and Productivity Gains by Industry



*Notes:* Data source are the November 2024 wave of the RPS, ages 18-64 (the August 2024 wave did not ask about time savings). The sample is employed individuals (RPS,  $N = 3473$ ). Labor productivity is calculated as real value added (Bureau of Economic Analysis) divided by employment (Bureau of Labor Statistics) in each sector. We exclude Agriculture because the productivity cannot be constructed with the standard BLS data, and Mining, Utilities, Wholesale Trade, and Management industries due to low sample size. Panel (a) plots industry-wide time savings from genAI against industry-wide hours worked last week using genAI, with both variables expressed relative to total industry hours worked last week. Panel (b) plots estimated productivity gains, as defined in Equation (4), against the industry-wide hours using genAI last week relative to total industry hours worked last week. When calculating the mean share of last week's work hours spent using genAI, we weight observations by their last week's work hours to obtain an aggregate estimate of genAI usage as a share of hours worked. See text for details on the calculation of estimated productivity gains. Panels (c) and (d) plot estimated changes in effective labor and productivity against annualized labor productivity growth in the US over the years 2022 - 2024, detrended by subtracting the annualized growth predicted by the trend over the period 2015-2019 from the actual growth rate. To minimize the impact of outliers, panels (b), (c), and (d), which weight time savings using wages, exclude observations with (i) hourly wages below the national minimum wage of \$7.25 or (ii) weekly earnings above \$10,000.

We model aggregate or industry output with a Cobb-Douglas function:

$$Y = AK^\alpha L^{1-\alpha}, \quad (1)$$

$$L = \sum_{i=1}^N \ell_i e_i, \quad (2)$$

where  $Y$  is output,  $K$  is capital,  $\ell_i$  is hours worked by worker  $i$ ,  $e_i$  is the efficiency of worker  $i$ , and  $L$  is the effective quantity of labor. If worker  $i$  saves  $s_i$  hours per week due to genAI and spends this time on additional production within their job, their effective labor becomes  $\ell_i + s_i$  and the change in the effective quantity of labor is

$$\Delta L = \sum_{i=1}^N s_i e_i. \quad (3)$$

Assuming a competitive labor market and constant TFP, capital, and hours worked, Appendix D.1 shows that we can write the approximate percent change in output from genAI as:

$$\frac{\Delta Y}{Y} \approx \underbrace{(1 - \alpha)}_{\text{labor cost share}} \times \underbrace{\frac{\sum_i s_i \tilde{w}_i}{\sum_i \ell_i \tilde{w}_i}}_{\Delta \% \text{effective labor}} \quad (4)$$

where  $\tilde{w}_i$  is worker  $i$ 's wage relative to mean wages. This equation states that the percent change in output due to genAI is the ratio of mean time savings to mean hours worked, weighted by worker's wages, and scaled by labor's share of production. Information on time savings and wages is available in the November 2024 RPS.

Figure 9a plots genAI use against time savings at the industry level using RPS data. We observe considerable variation in genAI use across industries, which is strongly correlated with genAI time savings. Information Services and Finance have the highest genAI use (9.4% and 9.6%, respectively) and time savings (2.7% and 2.3%, respectively), while Accommodation and Food Services has the lowest use (1.5%) and time savings (0.6%). The slope of the dashed OLS line is 0.21, indicating that a 10 percentage point increase in genAI use is associated with a 2.1% rise in time savings. The relationship is approximately linear; in particular, we do not observe a plateauing of time savings at higher rates of adoption.

Next, Figure 9b translates time savings into productivity gains using equation (4). The impact of time savings on productivity depends on two factors: (i) the share of labor in production costs, and (ii) the wages of genAI users relative to non-users within the industry. (Appendix Table D.1 lists the labor shares used for each industry.)

The implied productivity gains range from a low of 0.1% in Real Estate to a high of 1.7%

in Professional, Scientific, and Technical. The correlation between productivity gains and AI usage by industry (0.65) is smaller than for the time-savings and industry (0.93). This is primarily due to variation in labor shares across industries. For example, Information Services has high genAI use and time savings, but a low labor share (0.39).

To what extent are these predicted productivity gains evident in economic data? Figure 9c compares changes in effective labor supply in the RPS with annualized labor productivity growth (i.e., the percent change in real value added per worker) from 2022 – 2024. To adjust for trends in TFP growth prior to the Covid-19 pandemic, we detrend annualized labor productivity by subtracting the annualized growth predicted by the trend over the period 2015-2019 from the actual growth rate. We find a positive correlation of 0.39. The power of this analysis is limited by three factors. First, our predicted changes in effective labor supply are modest relative to observed changes in labor productivity, which can be affected by many factors other than genAI. Second, the change in labor productivity over this period may partly reflect recovery from the Covid-19 pandemic: in particular, both Retail and Transportation were disproportionately negatively impacted by demand shifts during the pandemic and exhibit very high productivity growth from 2022-2024 relative to trend. Third, our sample size includes only 15 industries with sufficient sample size.

Figure 9d compares changes in predicted productivity gains in the model with annualized labor productivity growth. In this comparison the correlation declines to effectively zero. Mechanically, the correlation declines because some industries with high effective labor gains and high observed productivity gains have a small labor share, which dampens the predicted productivity gain in the model. One potential explanation for the decrease in the correlation is that industry adoption of genAI may be correlated with worker adoption. If industries with large increases in effective labor supply also experienced productivity gains due to firm-level adoption, then our model, which only captures changes in individual workers’ effective labor supply, would under-predict total productivity gains in those industries.

#### 5.4 Aggregate Productivity Gains and a Comparison of Micro and Macro Estimates

We can apply this same procedure to estimate aggregate productivity gains due to genAI. In particular, we estimate that on the aggregate level genAI currently increases the aggregate effective labor supply by 2.1%. Using an AI-exposure-adjusted labor share of 0.57, following Acemoglu (2024), we estimate a potential productivity gain of 1.2%.

To compare our estimate of the aggregate productivity gain from genAI with experimental estimates from the literature, we reformulate the model to express time savings as a linear

function of time spent using genAI,  $s_i = \gamma u_i$ :

$$\Delta L = \sum_{i=1}^N \gamma u_i e_i. \quad (5)$$

In this expression,  $u_i$  is weekly hours spent using genAI by worker  $i$  and  $\gamma$  is the productivity gain associated with one hour of genAI use.

Given our estimates of genAI use, Appendix D.1 shows that the value of  $\gamma$  that would generate a 2.1% increase in aggregate effective labor supply can be written as:

$$\gamma = 2.1\% \cdot \left( \frac{\sum_i \tilde{w}_i l_i}{\sum_i \tilde{w}_i u_i} \right) \quad (6)$$

As discussed in Section 5.1, RPS data provide a lower and upper bound for  $l_i$ ,  $\underline{l}_i$  and  $\bar{l}_i$ . Evaluating (6) using the midpoint of these bounds,  $l_i \equiv (\underline{l}_i + \bar{l}_i)/2$  yields a wage-weighted share of total work hours spent using genAI of 5.7%, which yields  $\gamma = 0.30$ . This implies that each hour spent using genAI increases the worker’s productivity for that hour by 30%. This is similar in magnitude to the average productivity gain of 27% from several randomized experiments of genAI usage (Cui et al., 2024; Dell’Acqua et al., 2023; Noy and Zhang, 2023; Peng et al., 2023).

Our estimated aggregate productivity gain from genAI (1.2%) exceeds the 0.7% estimate by Acemoglu (2024) based on a similar framework. In our framework and in Acemoglu (2024), the aggregate productivity gain is the product of three numbers: the AI-exposure-adjusted labor share of GDP, the wage-weighted share of labor that is supplied by genAI, and the productivity gain from genAI. We use the same labor share as Acemoglu (2024). Our survey data imply a wage-weighted genAI adoption share of 7.0%, while Acemoglu (2024) assumes a slightly lower wage-weighted adoption share of 4.6% based on task-exposure predictions. As discussed above, we estimate a productivity gain from genAI of 30%; Acemoglu (2024) assumes a productivity gain from genAI of 27% based on micro experimental evidence. Adjusting our productivity value  $\gamma$  from 30% to 27% lowers our estimated aggregate productivity gain to 1.1%; the remaining 0.4 percentage point gap is due to our higher adoption estimate.

We conclude with two points about our estimate of aggregate productivity gains. First, Acemoglu (2024) assumes these productivity gains take a decade to materialize. In contrast, our estimates reflect current adoption and time savings, implying a faster pace of diffusion. Second, our estimate does not capture potential productivity gains from using genAI to reorganize or completely automate production processes, which could further increase output per worker. With previous technologies, these more systematic gains occurred with substantial time lags, implying that this omission may be modest in the short run (Bresnahan and Trajtenberg, 1995; Brynjolfsson and Hitt, 2000; David, 1990; Jovanovic and Rousseau, 2005).

## 6 Conclusion

Predictions about genAI’s economic impact vary widely, driven by uncertainty over the share of work it can assist and the size of its productivity potential. This paper presents evidence from the first nationally representative U.S. surveys on generative AI use at work and home. We find that a substantial share of individuals already use genAI. Thus far, the pace of overall adoption is rapid compared with past ICT technologies. At work, genAI adoption has followed a similar pace to PCs and is far ahead of genAI adoption by firms. We document wide variation in genAI use across demographic groups and occupations, and provide evidence that both task composition and firm policies appear to play an important role in workers’ adoption choices. Although the intensity of genAI use varies widely, genAI already assists a noticeable share of total work hours, suggesting the potential for meaningful productivity gains.

Our findings help lay the groundwork for understanding the labor market implications of genAI. They can also inform responses by managers and policymakers along several dimensions, such as data center and electricity needs, workforce training, and competitive dynamics.

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# The Rapid Adoption of Generative AI

## ONLINE APPENDIX

## A RPS: Measurement and Definitions

### A.1 Sample Restrictions

The Qualtrics panel includes about 15 million members and is not a random sample of the U.S. population. However, researchers can instruct Qualtrics to target survey invitations to specific demographic groups. The RPS sample was designed to be nationally representative of the U.S. across gender, age, education, household income, and other demographic characteristics. Individuals who take the survey too fast, i.e. in less than 50% of the median time in a soft launch, or who do not state that they will provide their best answers, are automatically dropped from the sample.<sup>1</sup> Finally, once fielding is completed Qualtrics staff goes over all responses and filters out any that look suspicious. This is typically the case for around 10% of the responses. We also screened manually out responses whose responses to the free text question about the industry appear to be generated by a bot taking the survey. This was the case for less than 10 responses in each survey.

The pilot survey in June 2024 received 2,551 responses. Our full surveys in August and November 2024 received 5,014 and 5,329 responses, respectively. In each survey, we dropped less than 40 respondents because they either reported their industry and/or occupation as military or because they reported being employed but also reported being homemakers, retired, or unemployed as their occupation. Table A.1 show the raw sample composition for the pooled August and November 2024 surveys after these observations were dropped.

We also include occupation in our weighting scheme. This requires us to drop another 108, 112, and 113 observations due to missing occupation codes for the June, August, and November surveys, respectively. We drop another 70, 154, and 145 employed respondents after constructing weights because we lack information on their Generative AI use at work last week. All of these dropped respondents were classified as “employed, absent from work last week” as they by construction cannot have used Generative AI last week. Accounting for all individuals dropped from our analysis, this leaves us with a final sample size of 94.7%, 96.8%, and 96.9% of the initially collected responses for the June and August surveys, respectively.

Table A.1 compares the sample composition between the CPS and RPS along the demographics targeted in the sampling procedure for our main surveys (columns 1 and 2). The most notable discrepancies are that individuals aged 18 to 24 and with no more than a high school

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<sup>1</sup>The exact phrasing of the screener question is: “We care about the quality of our survey data and hope to receive the most accurate measure of your opinions, so it is important to us that you thoughtfully provide your best answer to each question in the survey. Do you commit to providing your thoughtful and honest answers to the questions in this survey?” with the following three answer options (1) I will provide my best answers, (2) I will not provide my best answers, and (3) I can’t promise either way. According to Qualtrics staff, this question provides the best results in terms of screening out respondents.

degree are underrepresented in the RPS relative to the CPS, while individuals with household income of \$50,000 or less are overrepresented. The bottom panel of Table A.1 compares employment status in the CPS and RPS, statistics that have not been targeted in the sampling procedure. Employment rates are very similar across the two surveys, although individuals classified as unemployed according to the CPS definition are somewhat overrepresented in the RPS.

## A.2 Weighting

As described in the body of the paper, we asked Qualtrics to administer the survey to a sample of respondents who match the U.S. population along a few broad demographic characteristics: gender, five age bins (18-24, 25-34, 35-44, 45-54, 55-64), race and ethnicity (non-Hispanic White, non-Hispanic Black, Hispanic, other), education (high school or less, some college or associate degree, bachelor’s degree or more), marital status (married or not), number of children in the household (0, 1, 2, 3 or more), three income bins for household income over the last 12 months (<\$50k, \$50k-100k, >\$100k), and four Census regions. Table A.1 compare the sample composition between the pooled August and November RPS and the CPS along the demographics targeted in the sampling procedure for our main survey. The tables look very similarly for each survey wave separately, including the June pilot, and are available upon request. As discussed in the paper, we also compare at the bottom of the table employment status in the CPS and RPS, as well as the demographic composition for the employed. None of this latter sets of moments were targeted during the sampling of survey respondents.

Using the iterative proportional fitting (raking) algorithm of Deming and Stephan (1940), we construct sampling weights to ensure the RPS matches the CPS sample proportions for the same set of demographic characteristics included in the Qualtrics sampling targets for the overall sample, i.e., independent of employment status. However, we use more disaggregated categories for education and marital status, and we interact all categories with gender. In particular, for education, we distinguish between less than high school, high school graduate or equivalent, some college but no degree, associate degree, bachelor’s degree, and graduate degree. For marital status, we distinguish between married + spouse present, divorced, never married, and "other." We also condition on relationship status (spouse living in the same household, partner living in the same household, other).

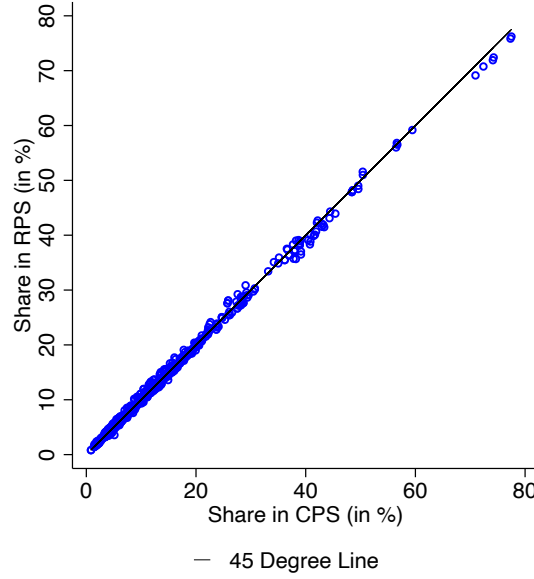
In addition, our sampling weights replicate the employed-at-work rates, the employment rates, and the labor force participation rates in each of the subsequent months. We match these key labor market statistics not only in the aggregate but also conditional on demographic characteristics. More specifically, we match the employed-at-work rate, the employment rate, and the labor force participation rate for the current month by gender, age (18-24, 25-34, 35-

Table A.1: Sample Composition in the Pooled August and November 2024 CPS and RPS

|                                           | <i>Everyone</i> |       | <i>Employed</i> |      |
|-------------------------------------------|-----------------|-------|-----------------|------|
|                                           | CPS             | RPS   | CPS             | RPS  |
|                                           | (1)             | (2)   | (3)             | (4)  |
| <i>Gender: Women</i>                      | 50.4            | 52.2  | 47.3            | 47.5 |
| <i>Age</i>                                |                 |       |                 |      |
| 18-24                                     | 15.0            | 11.7  | 12.0            | 12.0 |
| 25-34                                     | 22.2            | 22.0  | 23.9            | 23.8 |
| 35-44                                     | 22.2            | 23.8  | 24.4            | 25.0 |
| 45-54                                     | 20.1            | 21.2  | 21.7            | 21.7 |
| 55-64                                     | 20.6            | 21.2  | 17.9            | 17.5 |
| <i>Race/Ethnicity</i>                     |                 |       |                 |      |
| Non-hispanic White                        | 56.5            | 56.4  | 57.8            | 57.8 |
| Non-hispanic Black                        | 12.9            | 13.2  | 12.1            | 12.3 |
| Hispanic                                  | 20.5            | 20.4  | 20.2            | 19.8 |
| Other                                     | 10.1            | 10.0  | 9.9             | 10.1 |
| <i>Education</i>                          |                 |       |                 |      |
| Highschool or less                        | 37.1            | 32.9  | 32.8            | 27.1 |
| Some college/Associate's degree           | 25.8            | 27.6  | 25.2            | 27.6 |
| Bachelor's or Graduate degree             | 37.1            | 39.5  | 42.0            | 45.3 |
| <i>Marital Status: Married</i>            | 50.1            | 49.0  | 52.8            | 52.4 |
| <i>Number of children</i>                 |                 |       |                 |      |
| 0                                         | 58.1            | 55.7  | 57.2            | 52.8 |
| 1                                         | 17.8            | 19.7  | 18.1            | 21.2 |
| 2                                         | 14.7            | 16.5  | 15.5            | 18.4 |
| 3+                                        | 9.3             | 8.1   | 9.2             | 7.6  |
| <i>Household Income in Last 12 Months</i> |                 |       |                 |      |
| \$0-\$50,000                              | 26.1            | 31.4  | 19.9            | 22.6 |
| \$50,000-\$100,000                        | 29.9            | 29.5  | 30.7            | 32.1 |
| \$100,000+                                | 44.0            | 39.1  | 49.5            | 45.3 |
| <i>Region</i>                             |                 |       |                 |      |
| Northeast                                 | 17.0            | 18.3  | 16.9            | 18.8 |
| Midwest                                   | 20.3            | 19.4  | 21.1            | 19.2 |
| South                                     | 38.8            | 38.1  | 38.1            | 37.9 |
| West                                      | 23.8            | 24.2  | 23.9            | 24.2 |
| <i>Employment Status</i>                  |                 |       |                 |      |
| Employed, at work last week               | 71.7            | 69.8  |                 |      |
| Employed, absent from work last week      | 2.5             | 2.9   |                 |      |
| Unemployed                                | 3.2             | 8.0   |                 |      |
| Not in the labor force                    | 22.6            | 19.3  |                 |      |
| <i>Observations</i>                       | 115477          | 10241 | 85546           | 7450 |

*Notes:* Column 1 reports the sample composition in the pooled August and November 2024 Current Population Survey (CPS) for the variables targeted by Qualtrics in the sampling procedure. The employment status was the only variable not targeted. Column 2 reports the sample composition in the pooled August and November 2024 Real-Time Population Survey (RPS). The sample in both data sets is restricted to the civilian population ages 18-64. Columns 3 and 4 report the same outcomes for the employed (at work and absent from work last week).

FIGURE A.1: Sample Composition in the Weighted RPS vs. CPS for August and November 2024



*Notes:* The figure shows for all statistics used in the weighting scheme, the weighted fraction of individuals with the respective characteristics in the RPS (on the y-axis) and the CPS (on the x-axis).

44, 45-54, 55-64), race and ethnicity (non-Hispanic White, non-Hispanic Black, Hispanic, all other racial and ethnic groups), education (high school or less, some college or associate degree, bachelor's degree or more), marital status (married + spouse present, never married, other), relationship status (spouse living in the same household, partner living in the same household, other), presence of children in the household (yes or no), household income over the last 12 months (<\$50k, \$50k-100k, >\$100k), and region (Midwest, Northeast, South, and West using the Census definition). These groupings were chosen to ensure that each cell size is at least 30. We also include 2-digit occupation codes in our weighting scheme. Among the 22 occupations, we merge several occupations to ensure a sufficient sample size. In particular, for the June 2024 survey, we merge a) "Architecture and Engineering Occupations" and "Life, Physical, and Social Science Occupations", b) "Community and Social Service Occupations" and "Legal Occupations", c) "Healthcare Support Occupations" and "Protective Service Occupations", and d) "Farming, Fishing, and Forestry Occupations" and "Construction and Extraction Occupations." For the August and November 2024 survey, we proceeded similarly but, due to the larger sample size, did not need to merge "Architecture and Engineering Occupations" and "Life, Physical, and Social Science Occupations."

Including all the interaction terms, we have a total of 48 statistics (e.g., gender is one, and gender x education is another) which we weight on, with a combined 295 categories (e.g., gender has two categories, and gender x education has  $2 \times 6 = 12$  categories; for the June 2024 we only have 294 categories because of the additional occupational grouping). To visualize the goodness

of fit, we plot in Figure A.1 for all statistics used in the weighting scheme, the weighted fraction of individuals with the respective characteristics in the RPS (on the y-axis) and the CPS (on the x-axis) for August and November 2024. The lack of sizeable deviations from the 45-degree line demonstrates how well the weighting procedure works. The results are similar for each survey wave separately, including the June pilot, and are available upon request.

### A.2.1 Weighting Adjustment Based on Awareness

In addition to the raking procedure described above to ensure representativeness along demographic characteristics, our weighting scheme incorporates results from a new experimental survey that tested alternative question sequences for measuring genAI awareness and use. This experiment revealed that our original question sequence likely underestimated genAI awareness. In response, we applied a reweighting adjustment to our 2024 survey waves to align them with the higher awareness rates observed under the alternative design. This raises reported usage levels modestly (by about 5–6 percentage points) without altering any qualitative patterns in the results.

**Details.** In survey waves prior to August 2025, respondents were first asked whether they had heard of generative AI and, conditional on having heard of it, about their usage. To assess whether this structure induced systematic differences in reporting, we introduced a randomized experiment in a survey conducted in August 2025. Respondents were randomly assigned to one of two groups. Group 1 received the original sequential flow, where they were first asked whether they had heard of genAI, and then whether they used it. Alternatively, Group 2 was first asked whether they used genAI, and if they did not use it they were asked why, with one option being that they had not heard of it.

**Measurement differences across groups.** We found that a larger share of respondents reported having heard of genAI in Group 2. Specifically, 79.6 percent of Group 1 reported having heard of genAI, compared with 88.4 percent of Group 2. Across the two groups, a similar share of respondents reported using genAI conditional on having heard of genAI, suggesting that differential awareness reporting was evenly split between genAI users and non-users. The higher awareness rate in Group 2, combined with similar usage rates conditional on awareness between Groups 1 and 2, translated into higher overall usage rates in Group 2. Similar patterns held across employment status and for usage at home and at work. These results indicate that the initial awareness question (for Group 1) generated a downward bias in reported usage rates. One possible explanation for this pattern is that some respondents strategically reported that they had not heard of genAI because they guessed they would receive fewer questions with that answer.

**Reweighting strategy.** In the August 2025 wave, we can adjust for this bias by reweighting Group 1 respondents to match Group 2 usage rates. We constructed adjustment weights across gender, age, education, employment status, and usage (at work and at home) using a raking algorithm that increases the weight of Group 1 users relative to non-users while leaving Group 2 weights unchanged. To maintain coverage across demographic–usage cells, we did not incorporate intensity of usage into the adjustment. (Alternatively, we find that weighting on intensity of genAI interacted with only one demographic characteristics leads to very similar results.)

We then apply this adjustment to the 2024 survey waves used in this paper. Since all respondents in these prior waves received the question sequence of Group 1, we apply the adjustment to all respondents in these waves. Furthermore, the share of respondents stating they have heard of genAI (prior to our weighting adjustment) shows no trend between the surveys prior to the August 2025 survey and is very similar to Group 1 in the August 2025 survey (79.6% in vs. 79.4% in the August 2024 survey. This stability suggests that the bias in August 2025 likely existed consistently across earlier waves as well.

### A.3 Validation Checks

To illustrate how the RPS and CPS compare for characteristics that are not part of the sampling scheme and the effect of weighting, panels (a) and (b) of Figure A.2 compare the usual weekly earnings distribution in the RPS and CPS in the pooled August and November 2024 waves, with and without weights, respectively. The unweighted distributions are already similar, and the weights improve the fit further.

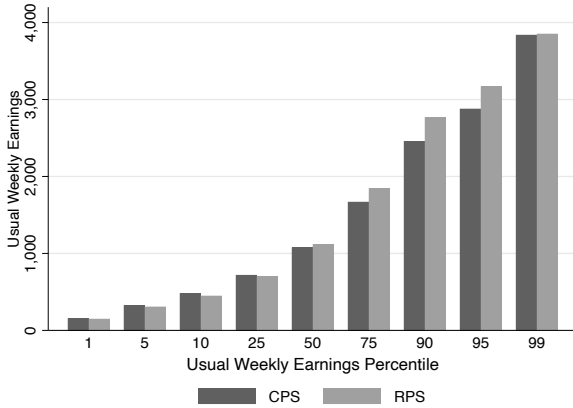
Panels (c) and (d) of Figure A.2 compare occupation shares in the RPS and the CPS, unweighted and weighted respectively. The unweighted correlation is 0.88, with Management as the only major outlier. Applying the weights corrects this imbalance and mechanically increases the correlation to 1.

Figure A.3 compare the industry shares in the RPS and CPS in the pooled August and November 2024 waves, with and without weights, as well as college major shares in the RPS and 2022 American Community Survey (ACS). The unweighted correlations of the industry shares is 0.89, comparable to those for occupations. The correlation of college majors in the RPS and the ACS is 0.83. The major with the highest share relative to the ACS is Computers, Mathematics, & Statistics, while the major with the lowest share relative to the ACS is Science and Engineering Related Fields. Weighting the data marginally improves the fit between the RPS and the two other data sets.

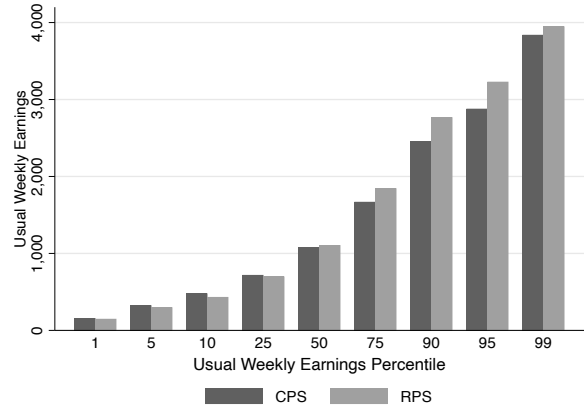
All four variables align closely across survey waves, including the June pilot, and are avail-

FIGURE A.2: Validation Checks – Usual Weekly Earnings and Occupation Shares

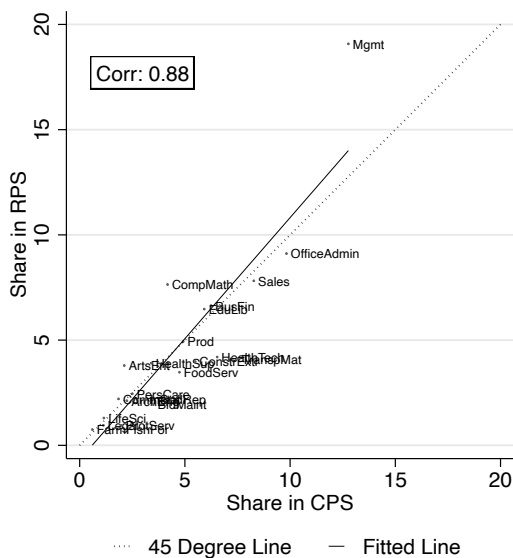
(a) Weekly Earnings Percentiles: Unweighted



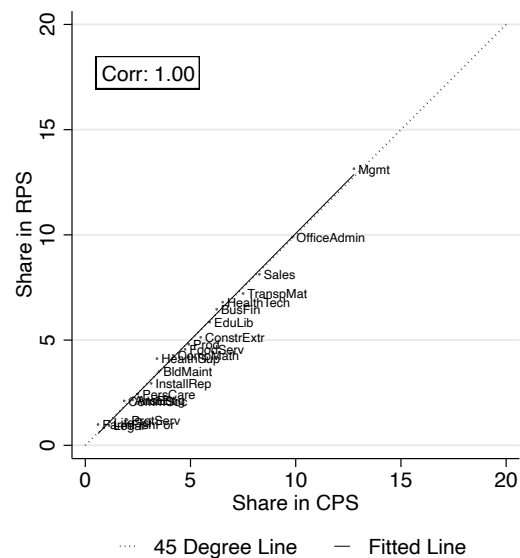
(b) Weekly Earnings Percentiles: Weighted



(c) Occupation Shares: Unweighted

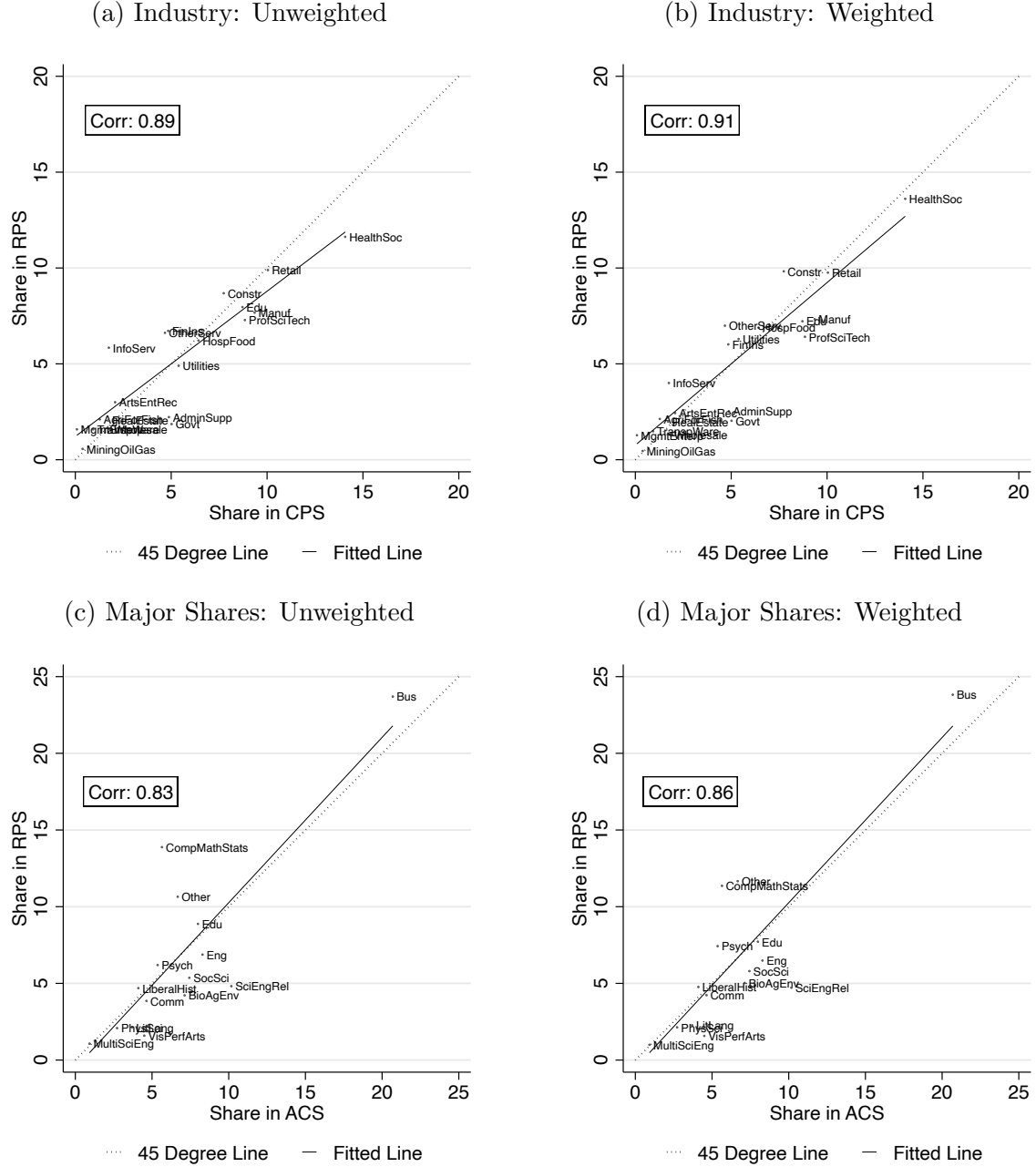


(d) Occupation Shares: Weighted



*Notes:* Figures on the left use unweighted RPS data, figures on the right use weighted RPS data. We use the same sample of RPS respondents in both figures. All figures use weighted CPS data. Data samples for the weekly earnings figures are employees ages 18-64 in the pooled August and November 2024 RPS and CPS-ORG with weekly earnings below the CPS topcode of \$4140 and an implied hourly wage of at least the federal minimum wage of \$7.25. Since the CPS topcode varies across month, we apply the lower topcode in both months. Sample sizes for the RPS and CPS are 4778 and 12133, respectively. Data samples for the occupation comparison are employed respondents ages 18-64 in the pooled August and November 2024 RPS and CPS with sample sizes of 6935 and 85546, respectively.

FIGURE A.3: Validation Checks – Industry and Major Shares



*Notes:* Figures on the left use unweighted RPS data, figures on the right use weighted RPS data. We use the same sample of RPS respondents in both figures. All figures use weighted CPS and ACS data, respectively. Data samples for the industry comparison are employed respondents ages 18-64 in the pooled August and November 2024 RPS and CPS with sample sizes of 6935 and 85546, respectively. Data samples for the college major comparison are respondents with a college degree ages 18-64 in the pooled August and November 2024 RPS with sample sizes of 3840 and 85546 in the 2022 ACS, respectively.

able upon request. The only exception is industry composition in the June 2024 survey (excluded from the main text), where the correlation is 0.74. This results from a higher “Other Services” share in the RPS than in the CPS, a recurring issue in past RPS waves Bick and Blandin, 2023. We attribute this to respondents selecting “Other” when unsure of their industry, unlike in the CPS, where professional coders classify responses. To address this, starting in the August 2024 survey, we prompted those selecting “Other Services” with the following question: “We would like to know some more details about what kind of business or industry this job is. Please include the main activity, product, or service provided at the location where you are employed. (For example: elementary school, residential construction).” We provided an open text field for respondents to type their answers. These responses were routed to an industry and occupational coder at the National Institute for Occupational Safety and Health (NIOSH) (<https://csams.cdc.gov/nioccs/SingleCoding.aspx>). Based on the provided responses, the system suggested the top five 6-digit NAICS codes plus an “Other” option. We assigned industry based on their choice, with “Other” being classified as “Other Services - Other Personal Services.” Figure A.3 confirms this resolved the issue.

#### A.4 Definition of Industries and Occupations

Industries correspond to the 22 major industries in the NAICS: Agriculture, Forestry, Fishing, and Hunting (NAICS=11); Mining, Quarrying, and Oil and Gas Extraction (NAICS=21); Utilities (NAICS=22); Construction (NAICS=23); Manufacturing (NAICS=31-33); Wholesale Trade (NAICS=42); Retail Trade (NAICS=44-45); Transportation and Warehousing (NAICS=48-49); Information (NAICS=51); Finance and Insurance (NAICS=52); Real Estate and Rental and Leasing (NAICS=53); Professional, Scientific, and Technical Services (NAICS=54); Management of Companies and Enterprises (NAICS=55); Administrative and Support and Waste Management Services (NAICS=56); Educational Services (NAICS=61); Health Care and Social Assistance (NAICS=62); Arts, Entertainment, and Recreation (NAICS=71); Accommodation and Food Services (NAICS=72); Other Services (except Public Administration) (NAICS=81); Public Administration (NAICS=99).

The industry groupings used in the main text figures are: Agriculture / Extraction (sectors 11, 21), Construction (23), Manufacturing (31-33), Wholesale / Retail Trade (42, 44-45), Transportation / Utilities (22, 48-49), Info Services (51), Finance / Real Estate (52, 53), Professional / Business Services (54, 55, 56), Education / Health/ Social / Public Services (61, 62, 92), Leisure / Accommodation / Other (71, 72, 81).

Occupations correspond to the 22 major occupations in the SOC: Management Occupations (SOC=11); Business and Financial Operations Occupations (SOC=13); Computer and Mathematical Occupations (SOC=15); Architecture and Engineering Occupations (SOC=17); Life,

Physical, and Social Science Occupations (SOC=19); Community and Social Service Occupations (SOC=21); Legal Occupations (SOC=23); Educational Instruction and Library Occupations (SOC=25); Arts, Design, Entertainment, Sports, and Media Occupations (SOC=27); Healthcare Practitioners and Technical Occupations (SOC=29); Healthcare Support Occupations (SOC=31); Protective Service Occupations (SOC=33); Food Preparation and Serving Related Occupations (SOC=35); Building and Grounds Cleaning and Maintenance Occupations (SOC=37); Personal Care and Service Occupations (SOC=39); Sales and Related Occupations (SOC=41); Office and Administrative Support Occupations (SOC=43); Farming, Fishing, and Forestry Occupations (SOC=45); Construction and Extraction Occupations (SOC=47); Installation, Maintenance, and Repair Occupations (SOC=49); Production Occupations (SOC=51); Transportation and Material Moving Occupations (SOC=53).

The occupation groupings used in the main text figures are: Personal Services occupations combine SOC codes 31-39: Healthcare support, Protective services, Food preparation and serving, Cleaning and maintenance, and Personal care. Blue Collar occupations combine SOC codes 47-53: Construction, Extraction, Installation, Maintenance and Repair, Production, Transportation, and Moving.

We elicited respondents' job titles through a free text response with autocomplete suggestions covering over 40,000 occupations from O\*NET and the Occupational Outlook Handbook and then match them to Standard Occupation Classification (SOC) codes using a parsing algorithm that identifies occupations in 97 percent of cases. For job titles that do not exactly match a unique SOC code, we present respondents with a choice of probabilistic matches using the job title to SOC code matching algorithm developed by the National Institute for Occupational Safety and Health (Laughlin et al., [2024](#)).

## A.5 Reasons for Not Using Generative AI at Work

In the November 2024 wave, we asked people who know about genAI but do not use it at work the following question.

*You indicated that you do not use Generative AI for your job. Below is a list of common reasons why people do not use Generative AI for their job. Please select all that apply to you.*

The answer options were:

1. Generative AI cannot help with my job.
2. I am unsure how to use Generative AI effectively for my job.
3. My employer does not allow me to use Generative AI.

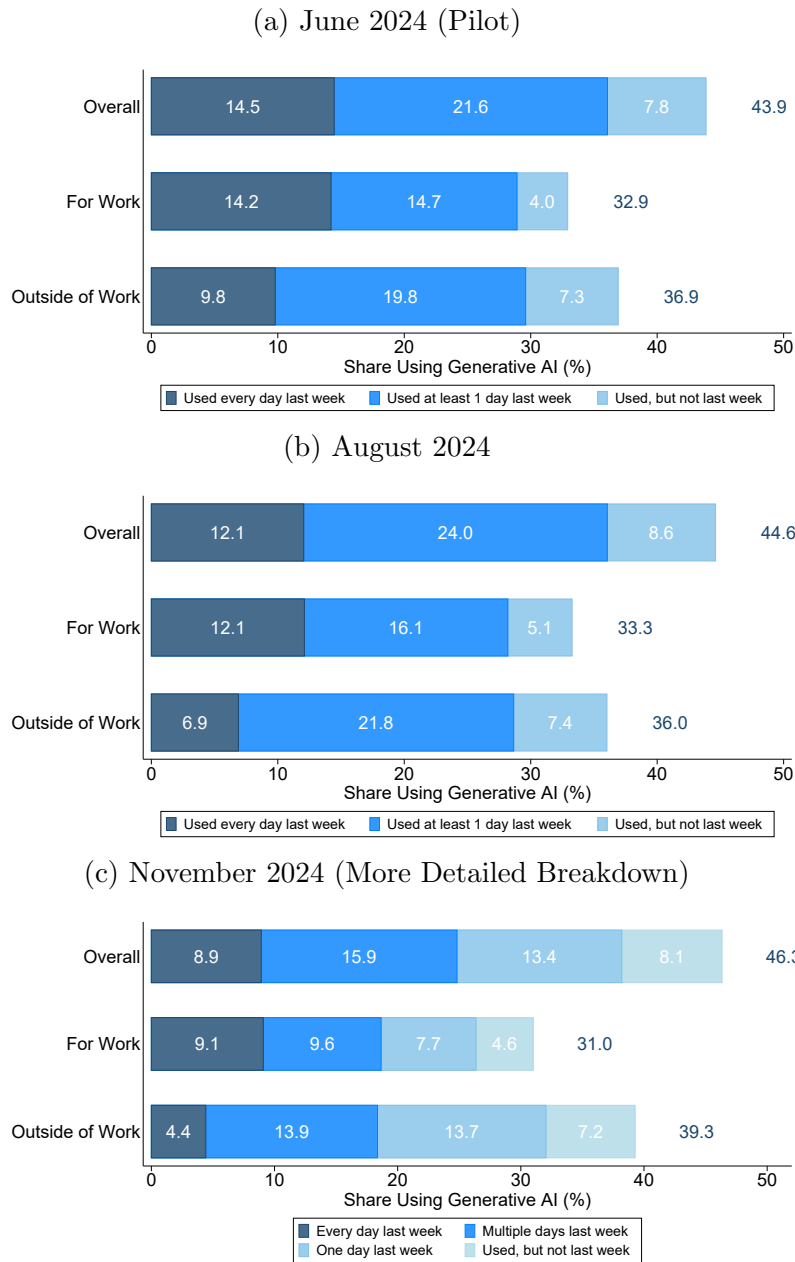
4. I am unsure whether my employer allows me to use Generative AI.
5. I feel pressure from coworkers to not use Generative AI.
6. I have privacy or security concerns.
7. I have ethical concerns.
8. I am skeptical about the quality and accuracy of AI-generated output.
9. I am worried that using Generative AI could eventually replace some or all parts of my job.
10. I am worried that using Generative AI would dilute my personal style or the creativity and authenticity of my work output.
11. Other reasons

We then followed up with the following question, displaying only the answer options selected in the previous question.

*You indicated that you not using Generative AI for your job for the following reasons. Please rank these tasks from “most important” (top) to “least important” (bottom) by dragging them.*

## B Comparison of Generative AI Use in the June (Pilot), August, and November 2024 Waves

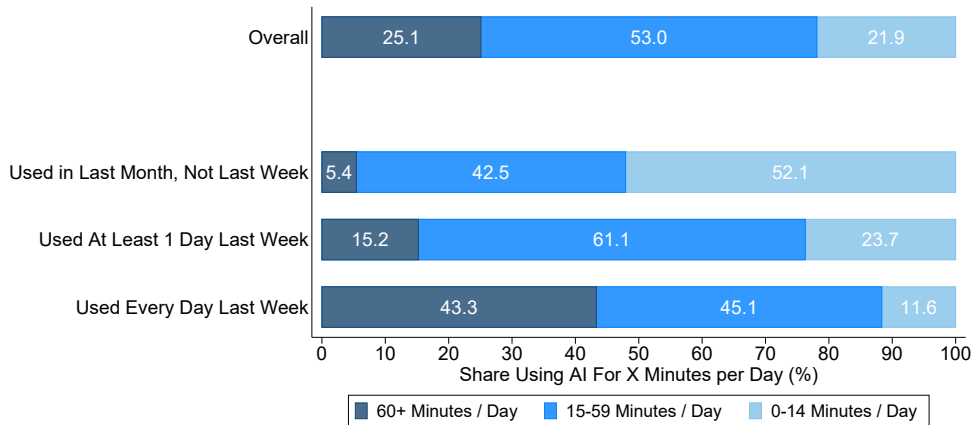
FIGURE B.1: Share of Working Age Adults Using Generative AI



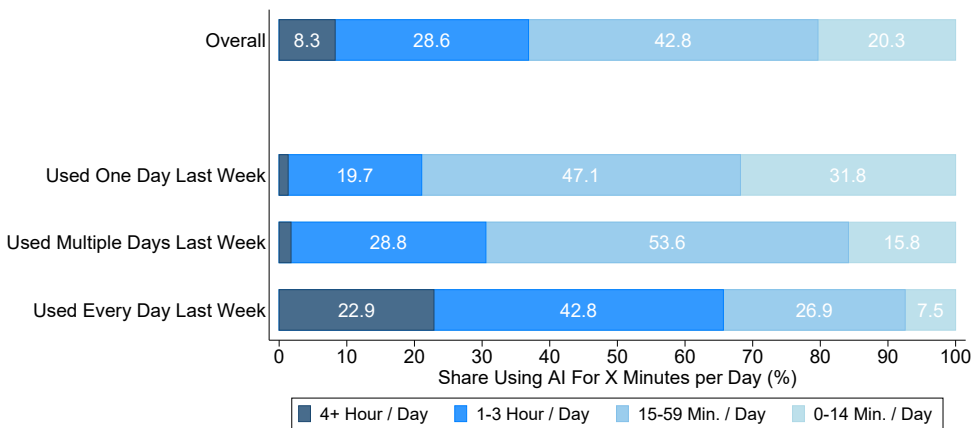
*Notes:* The figure shows the share of respondents who use genAI for work, outside of work, and overall (either for work or outside of work). Intensity of use is broken down into every day last week (dark blue), at least one day but not every day last week (medium blue), and not last week (light blue). Data source is the June 2024 wave of the RPS (Panel a), the August 2024 wave (Panel b), and the November 2024 wave (Panel c). The “For Work” samples are employed individuals (June,  $N = 1573$ ; August,  $N = 3238$ ; November,  $N = 3697$ ). The other bars include all respondents (June,  $N = 2351$ ; August,  $N = 4704$ ; November,  $N = 5022$ ).

FIGURE B.2: Intensity of genAI Use For Work

(a) August 2024



(b) November 2024 (More Detailed Breakdown)



*Notes:* The figure shows the distribution of daily time spent actively using genAI for work, among genAI users. In August, usage time is broken down into 0-14 minutes per day (light blue), 15-59 minutes per day (medium blue), and 60 or more minutes per day (dark blue). In November, usage time is broken down into 0-14 minutes per day (lightest blue), 15-59 minutes per day (light blue), 1-4 hours per day (medium blue), and 4 or more hours per day (dark blue). The “Overall” bar reflects the distribution among all genAI users. Data source is the August and November 2024 waves of the RPS, ages 18-64 (the June 2024 Pilot did not ask about time spent using genAI). The sample is employed respondents who use genAI for work. The sample for this figure is employed individuals with a valid occupation (August,  $N = 983$ ; November,  $N = 1000$ ).

## C SWAA

The Survey of Working Arrangements and Attitudes (SWAA) is a monthly cross-sectional survey of Americans aged 20 to 64 that has been running since May 2020. In December 2024, we incorporated a subset of our questions on genAI usage into the SWAA. Similar to the RPS, the SWAA is fielded online, though it is conducted by a different commercial survey provider (IncQuery) and targets a sample representative of the U.S. population aged 20–64 by age, sex, income, race, and region. Table C.1 shows the analogue of Table A.1 and compares the unweighted SWAA to the weighted CPS. In addition to the targeted variables by the SWAA, we also include live-in-partner (in lieu of marital status) and the number of children. Individual income in the last 12 months is not collected by the SWAA. Similar to the RPS, the SWAA matches the targeted (and non-targeted) quotas fairly closely and the unemployed are somewhat overrepresented. We note that the SWAA uses a single question to determine employment status, which differs from the more detailed protocol used in the CPS and RPS, and map the SWAA’s employment categories into the CPS definitions.<sup>2</sup>

### C.1 Weights

To maximize comparability between the RPS and SWAA, we construct our own set of weights rather than using the weights provided by the SWAA.<sup>3</sup> In particular, we use again the iterative proportional fitting (raking) algorithm of Deming and Stephan (1940). We include all the demographic variables shown in Table C.1 and also interact all of them with gender. The sampling weights further replicate the employed-at-work rate, the employment rate, and the labor force participation rate in the aggregate and interacted with the demographic variables used for overall weighting. To ensure a sufficient sample size for these latter interactions, we use a broader group for age (20-29, 30-39, 40-49, 50-64) and a dummy variable for whether there is a child in the house. We also include the broad occupation groups in our weighting scheme, and as for the RPS, without any interaction.

### C.2 Validation

Figure C.1 shows that similar to the RPS the occupation and industry distribution in SWAA closely aligns with CPS in December 2024.

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<sup>2</sup>In particular, the SWAA asks: “Last week what was your work status?” with the following answer options (a) Working for pay, whether on business premises or working from home, (b) Still employed and paid, but not working, (c) Unemployed, looking for work, (d) Unemployed, awaiting recall for my old job, and (e) Not working, and not looking for work.

<sup>3</sup>Notably, the employment-to-population ratio calculated using the SWAA’s provided weights is with 59% much lower than in the raw SWAA data (76.2%) and in the CPS (75.7%).

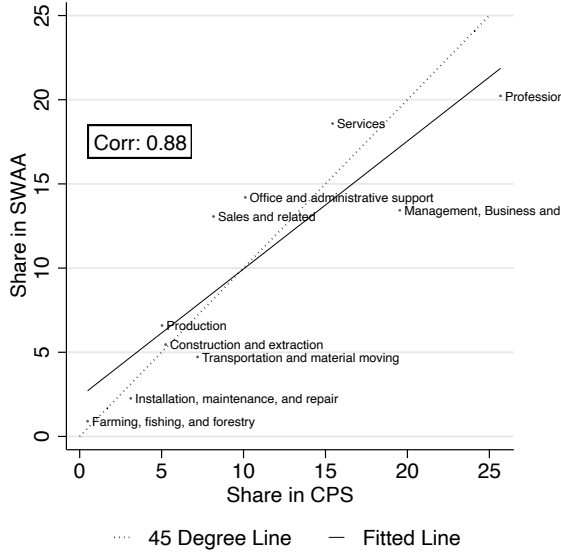
Table C.1: Sample Composition in the December 2024 CPS and SWAA

|                                      | <i>Everyone</i> |      | <i>Employed</i> |      |
|--------------------------------------|-----------------|------|-----------------|------|
|                                      | CPS             | SWAA | CPS             | SWAA |
|                                      | (1)             | (2)  | (3)             | (4)  |
| <i>Gender: Women*</i>                | 50.5            | 58.6 | 47.3            | 54.7 |
| <i>Age*</i>                          |                 |      |                 |      |
| 20-24                                | 11.2            | 6.8  | 9.9             | 7.2  |
| 25-34                                | 23.2            | 20.4 | 24.5            | 21.8 |
| 35-44                                | 23.2            | 26.1 | 25.0            | 28.8 |
| 45-54                                | 20.9            | 23.9 | 22.1            | 24.1 |
| 55-64                                | 21.5            | 22.7 | 18.5            | 18.2 |
| <i>Race/Ethnicity*</i>               |                 |      |                 |      |
| Non-hispanic White                   | 56.8            | 64.4 | 57.9            | 63.0 |
| Non-hispanic Black                   | 12.8            | 10.1 | 12.1            | 10.3 |
| Hispanic                             | 20.3            | 19.3 | 20.1            | 20.5 |
| Other                                | 10.1            | 6.2  | 9.9             | 6.2  |
| <i>Education*</i>                    |                 |      |                 |      |
| Highschool or less                   | 35.4            | 34.6 | 31.8            | 30.3 |
| Some college/Associate's degree      | 25.7            | 29.1 | 25.1            | 28.5 |
| Bachelor's or Graduate degree        | 38.9            | 36.3 | 43.1            | 41.2 |
| <i>Live-in-Partner</i>               | 60.1            | 55.4 | 62.7            | 55.5 |
| <i>Number of children</i>            |                 |      |                 |      |
| 0                                    | 58.1            | 56.4 | 57.0            | 53.3 |
| 1                                    | 17.6            | 21.3 | 18.0            | 23.1 |
| 2                                    | 14.8            | 15.6 | 15.7            | 16.9 |
| 3+                                   | 9.5             | 6.8  | 9.3             | 6.8  |
| <i>Region*</i>                       |                 |      |                 |      |
| Northeast                            | 17.0            | 18.5 | 17.0            | 19.6 |
| Midwest                              | 20.3            | 20.6 | 21.0            | 20.4 |
| South                                | 38.7            | 44.0 | 38.2            | 43.3 |
| West                                 | 24.0            | 16.9 | 23.8            | 16.7 |
| <i>Employment Status</i>             |                 |      |                 |      |
| Employed, at work last week          | 73.7            | 71.6 |                 |      |
| Employed, absent from work last week | 2.0             | 4.4  |                 |      |
| Unemployed                           | 2.8             | 9.7  |                 |      |
| Not in the labor force               | 21.5            | 14.3 |                 |      |
| <i>Observations</i>                  | 54949           | 4914 | 41481           | 3732 |

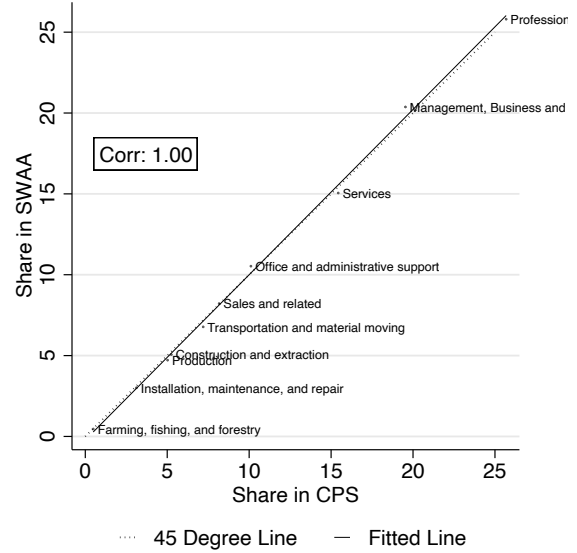
*Notes:* Column 1 reports the sample composition in the December 2024 Current Population Survey (CPS) for the variables most comparable to those targeted in the RPS in the sampling procedure. The SWAA targets only the variables marked with an asteriks. The employment status is neither targeted in the RPS nor the SWAA. Column 2 reports the sample composition in the December 2024 SWAA. The sample in both data sets is restricted to the civilian population ages 20-64. Columns 3 and 4 report the same outcomes for the employed (at work and absent from work last week).

FIGURE C.1: SWAA Validation Checks

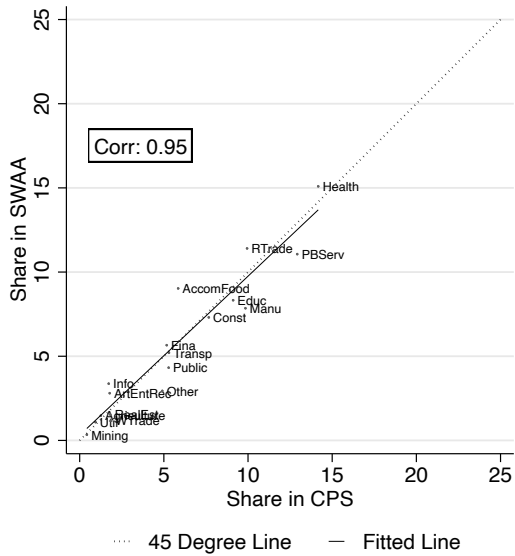
(a) Occupation Shares: Unweighted



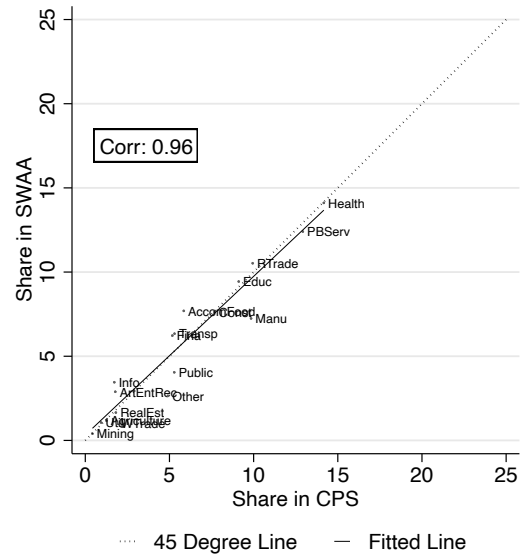
(b) Occupation Shares: Weighted



(c) Industry: Unweighted



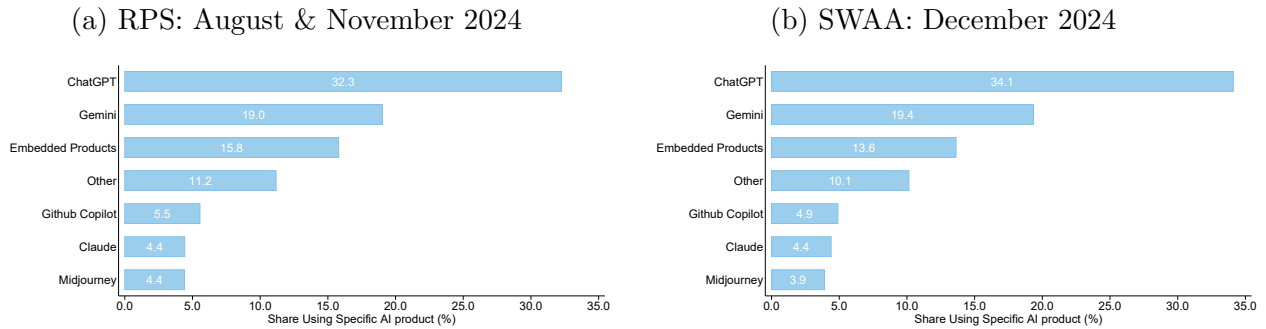
(d) Industry: Weighted



*Notes:* Figures on the left use unweighted SWAA data, figures on the right use weighted SWAA data using our own weights. All figures use weighted CPS data. Data samples are all employed respondents. Sample sizes for the December 2024 SWAA and CPS are 4419 and 43202, respectively.

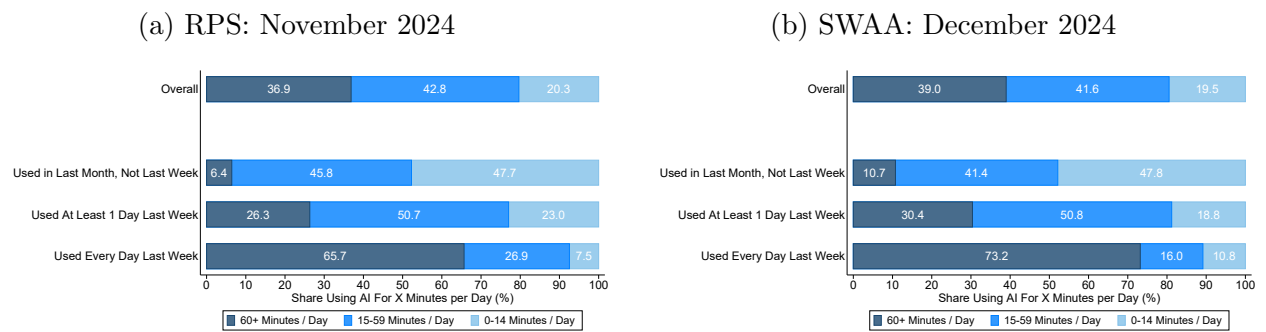
### C.3 Comparison of Generative AI Use in the RPS and SWAA

FIGURE C.2: Share of Respondents Using Specific Generative AI Products



*Notes:* The figure shows the share of respondents who report using particular genAI products. “Embedded products” are genAI features embedded within existing software, such as Microsoft Copilot. Data source is the August and November 2024 waves of the RPS, ages 18-64 ( $N = 9742$ ) and the December 2024 SWAA, ages 20-64 ( $N = 4698$ ). Individuals who report using multiple genAI products are reflected in multiple bars.

FIGURE C.3: Intensity of genAI Use For Work



*Notes:* The figure shows the distribution of daily time spent actively using genAI for work, among genAI users. Usage time is broken down into 0-14 minutes per day (light blue), 15-59 minutes per day (medium blue), and 60 or more minutes per day (dark blue). The “Overall” bar reflects the distribution among all genAI users. In Panel a, data source is the November 2024 waves of the RPS, ages 18-64. In Panel b, data source is the December 2024 SWAA, ages 20-64. The sample for both data sources is employed respondents who use genAI for work. (RPS,  $N = 1000$ ; SWAA,  $N = 861$ ).

## D Details on Time Savings and Productivity Calculations

### D.1 Derivations Using the Aggregate Production Model

#### The Aggregate Productivity Gain from Generative AI

We model the economy's aggregate output using a Cobb-Douglas aggregate production function where labor supply is perfectly substitutable across workers. The economy contains  $N$  workers. Let  $Y$  denote the aggregate output, and let  $L$  denote the aggregate supply of labor measured in efficiency units (going forward we refer to this as aggregate effective labor supply). The aggregate production function is given by:

$$Y = AK^\alpha L^{1-\alpha}, \quad (\text{D.1})$$

where  $A$  is TFP,  $K$  is the aggregate capital stock, and  $\alpha$  is the Cobb-Douglas share on capital. The aggregate effective supply of labor  $L$  is defined as the weighted sum of individual labor supplies:

$$L = \sum_{i=1}^N \ell_i e_i \quad (\text{D.2})$$

where  $\ell_i$  is the number of hours worked per week by worker  $i$ , and  $e_i$  is the efficiency units of labor supplied by worker  $i$ .

In a competitive labor market, each worker is paid their marginal product of labor:

$$w_i = (1 - \alpha)A(K/L)^\alpha e_i \quad (\text{D.3})$$

where  $w_i$  is the worker's hourly wage. Normalizing the mean efficiency unit to  $\bar{e} = 1$ , we have:

$$\tilde{w}_i \equiv \frac{w_i}{\bar{w}} = \frac{w_i}{(\sum_i w_i)/N} = \frac{(1 - \alpha)A(K/L)^\alpha e_i}{(\sum_i (1 - \alpha)A(K/L)^\alpha e_i)/N} = \frac{(1 - \alpha)A(K/L)^\alpha e_i}{(1 - \alpha)A(K/L)^\alpha \bar{e}} = e_i \quad (\text{D.4})$$

where  $\tilde{w}_i$  is worker  $i$ 's wage relative to mean wages  $\bar{w}$ .

Suppose that worker  $i$  saves  $s_i$  hours per week due to genAI and spends this time on additional production within their job. The effective weekly labor supply for worker  $i$  is then  $\ell_i + s_i$ . Substituting this into the expression for aggregate effective labor supply, we have:

$$L' = \sum_i (\ell_i + s_i) e_i = \sum_i \ell_i e_i + \sum_i s_i e_i. \quad (\text{D.5})$$

The change in aggregate effective labor supply attributable to genAI-induced time savings

is:

$$\Delta L \equiv L' - L = \sum_i s_i e_i = \sum_i s_i \tilde{w}_i \quad (\text{D.6})$$

Assuming no change in TFP, capital, and hours worked, the percent change in aggregate productivity from genAI equals the percent change in aggregate output. The approximate percent change in aggregate output from genAI is given by:

$$\frac{\Delta Y}{Y} \approx (1 - \alpha) \frac{\Delta L}{L} = \underbrace{(1 - \alpha)}_{\text{labor cost share}} \times \underbrace{\frac{\sum_i s_i \tilde{w}_i}{\sum_i \ell_i \tilde{w}_i}}_{\Delta \% \text{effective labor}} \quad (\text{D.7})$$

This equation states that the percent change in output due to genAI is the ratio of mean time savings to mean hours worked, weighted by worker's wages, and scaled by labor's share of production costs.

Using November 2024 RPS data on hourly wages  $w_i$ , weekly hours worked  $\ell_i$ , and weekly genAI time savings  $s_i$ , we use (D.7) to estimate that genAI currently increases aggregate effective labor supply by 2.1%. If we assume an AI-exposure-adjusted labor share of 0.57, following Acemoglu (2024), the implied potential productivity gain is 1.2%.

### Comparisons to Micro and Macro Estimates of GenAI Productivity Gains

How does our estimate of the aggregate productivity gain from genAI compare with experimental estimates from the literature? To answer this question, we reformulate the model to express time savings as a linear function of time spent using the technology:

$$L' = \sum_i (l_i + \gamma u_i) e_i = \sum_i l_i e_i + \gamma \sum_i u_i e_i \quad (\text{D.8})$$

In this expression,  $u_i$  is the weekly hours spent using genAI by worker  $i$ , and  $\gamma$  is the productivity gain associated with one hour of genAI use.

We can then write the percent increase in aggregate effective labor supply due to genAI as

$$\frac{\Delta L}{L} = \frac{\gamma \sum_i \tilde{w}_i u_i}{\sum_i \tilde{w}_i l_i} \quad (\text{D.9})$$

Next, we solve for the value of  $\gamma$  that would generate a 2.1 percent increase in aggregate effective labor supply given our estimates of genAI use:

$$\gamma = 2.1\% \cdot \left( \frac{\sum_i \tilde{w}_i l_i}{\sum_i \tilde{w}_i u_i} \right) \quad (\text{D.10})$$

Recall that for each worker in the RPS data we do not have a single estimate for  $l_i$ ; instead, we have a lower and upper bound,  $\underline{l}_i$  and  $\bar{l}_i$ , respectively. Evaluating (D.10) using the midpoint of these bounds,  $l_i \equiv (\underline{l}_i + \bar{l}_i)/2$ , yields a wage-weighted share of total work hours spent using genAI of 7.0%, which implies a productivity parameter value of  $\gamma = 0.30$ . This implies that each hour spent using genAI increases the worker’s productivity for that hour by 30%. This is similar in magnitude to the average value of 27% from several randomized experiments of genAI usage (Cui et al., 2024; Dell’Acqua et al., 2023; Noy and Zhang, 2023; Peng et al., 2023).

## D.2 Industry Labor Shares

Table D.1: Labor Shares by Industry

| Industry (NAICS2)                                                        | Labor Share |
|--------------------------------------------------------------------------|-------------|
| Mining, quarrying, and oil and gas extraction                            | 0.227       |
| Utilities                                                                | 0.237       |
| Construction                                                             | 0.717       |
| Manufacturing                                                            | 0.479       |
| Wholesale trade                                                          | 0.528       |
| Retail Trade                                                             | 0.583       |
| Transportation and Warehousing                                           | 0.660       |
| Information                                                              | 0.389       |
| Finance and Insurance                                                    | 0.547       |
| Real Estate and Rental and Leasing                                       | 0.070       |
| Professional, Scientific, and Technical Services                         | 0.759       |
| Management of Companies and Enterprises                                  | 0.887       |
| Administrative and Support and Waste Management and Remediation Services | 0.780       |
| Educational Services                                                     | 0.849       |
| Health Care and Social Assistance                                        | 0.846       |
| Arts, Entertainment, and Recreation                                      | 0.806       |
| Accommodation and Food Services                                          | 0.696       |
| Other Services (except Public Administration)                            | 0.837       |
| Public Administration                                                    | 0.503       |

Notes: Estimates are for the BLS in 2023 from Karabarbounis ([2024](#)).